R&D Support Project Proposal for SMEs Technology Development

❶ Project Overview

Project Name	Development of RAG-Based Business Specialization Real-Time Multilingual AI Translation Solution - Global Sales, Consultation, Translation, Document Automation Integrated Platform for SME Export Sales Innovation
Development Goal	Implementation of AI models specialized for business: Develop a RAG-based multilingual translation model trained on unique company knowledge (product names, technical specifications, prices, etc.) and an End-to-End (E2E) real-time streaming voice interpretation model (in more than 8 languages) with a delay of within 2 seconds. Automation of the entire export business cycle: Develop 5 types of export document auto-generation agents such as quotes and product pages. Unique regulatory certification formatting: Build a SaaS platform integrating complex regulatory certification document formatting modules such as eCTD, KC, CE, FCC to remove global entry barriers for SMEs and achieve sales innovation.
Development Method	[Core Differentiation Data] Establish an overwhelming entry barrier by conducting domain-specific fine-tuning using an exclusive corpus with 1.1 million sentence pairs (600,000 business + 500,000 regulatory certifications) that cannot be replicated in a short period. [AI Model Advancement] Combine RAG technology that injects company-specific knowledge into a hybrid language model based on NLLB/EXAONE to reduce hallucination phenomena by 70%. Secure simultaneous interpretation performance within a 2-second delay with InfiniSST and Whisper-based technology. [SaaS Platform Commercialization] Develop a cloud SaaS web, mobile, API integrated environment that allows demand companies to link and use their data within 3 days without IT personnel, utilizing the verified AI platform (Oz-i) infrastructure of the main organization.

Research Team Composition	[Main] PCN Co., Ltd. (Platform/Commercialization General Manager): 25 years of operation (annual sales of 27 billion won), an AI company with a GS-certified platform, possessing the Oz-i AI platform and extensive know-how in various R&D projects, ensures the construction of the SaaS platform, API integration, and cloud infrastructure commercialization. [Collaborator①] StunningBox Co., Ltd. (AI Model/Data Specialist): An AI translation data company with 15 years of accumulated professional datasets, responsible for AI translation fine-tuning, providing its core asset of an exclusive corpus, and multi-agent regulatory certification formatting technology. [Collaborator②] 3 Demanding Companies (Field Proof Verification): Sunlight Illucom Co., Ltd. (automotive parts), Genuine Grip Co., Ltd. (special shoes),-Grey Whale (K-Beauty), all actual top export award companies, provide proof data for three industries and quantitatively verify the introduction effects through A/B testing.
Expected Effects	[Economic and Industrial Impact] Enhance the export contract rate of demand companies by over 20% and drastically reduce the regulatory certification document processing period from the current 4–8 weeks to 1–2 weeks. By the fifth year of commercialization, create 22 billion won in sales (cumulative 68 billion won over 5 years), generate 51 new jobs, and plan to supply to over 500 SMEs by linking AI voucher and KOTRA. Achieve a cumulative 44 billion won export substitution effect over 5 years. [World-Class Technology Acquisition] Achieve translation accuracy with BLEU over 65, voice interpretation delay within 2 seconds, and document structure retention (TEDS) over 70%, to preempt the B2B domain-specialized translation and interpretation market that global big tech (Google, DeepL) could not solve.

2 Self-Assessment

Project Name	Global Sales, Consultation, Translation, Document Automation Integrated Platform for SME Export Sales Innovation	
Checklist	Applicable	Not Applicable
1. Does the current applicant (R&D organization, etc., hereinafter referred to as 'applicant') not meet the application qualifications specified in the business announcement, etc.?		✓
2. Are there any documents among the related documents stipulated by the business announcement, etc., that the current applicant has not submitted?		✓
3. Is the current applicant subject to national R&D participation restrictions?		✓
4. Does the applicant have a history of sanctions, such as restrictions on government R&D participation due to misuse of labor costs or fraudulent transactions?		✓
5. Is the current applicant a bankrupt, suspended, or closed company?		✓
6. Is the current applicant under tax delinquency disposition by the tax authorities for national tax, local tax, customs duties, etc., or registered with a comprehensive credit information concentration agency for fine delinquency, industrial accident/ employment insurance delinquency, etc.?		✓
7. Is the current applicant registered in the debtor list based on the Civil Execution Act or as a debtor with a comprehensive credit information concentration agency?		✓
8. Has the current applicant filed for the commencement of bankruptcy, rehabilitation procedures, or individual rehabilitation procedures in court?		✓
9. Based on the most recent year's financial statements, does the recent applicant have a debt ratio of over 1000% or is fully impaired in capital?		✓
10. As a result of its own inspection, does the application task fall under the security task classification criteria of Article 45 Paragraph 1 of the Enforcement Decree of the Innovation Act as a security task?		✓

※ In addition to the items specified in the self-assessment table above, support may be excluded or selection may be canceled depending on the review results of the application qualification criteria prescribed by the business announcement and related regulations ※ The criterion of 'current' in the checklist above is the point set for confirming the application qualification requirements in the business announcement, related regulations, etc., such as the application deadline ※ Each main and joint institution needs to submit

This applicant has confirmed the facts regarding the above items, and does not object to the possibility of support exclusion or selection cancellation if they differ from the facts.

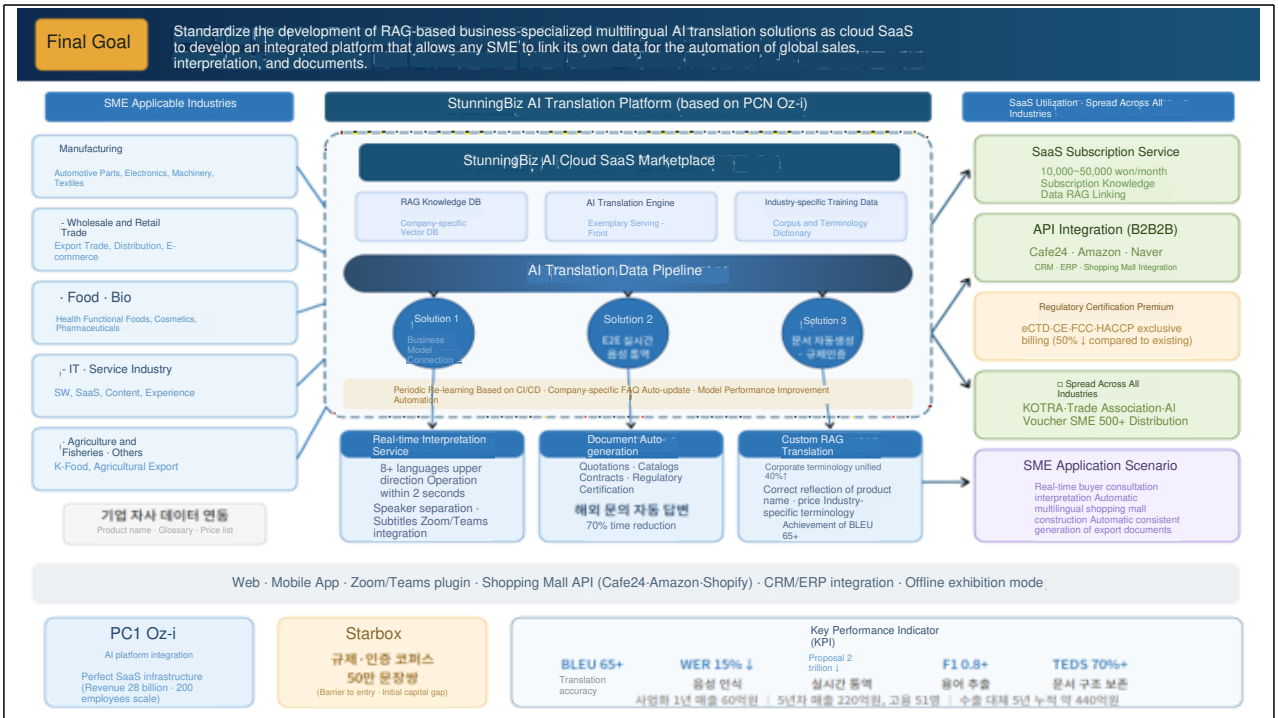
Year 2026 July Name

Main Institution Applicant: Sang-hyun Yeom



Main Institution Corporation Name: CN Co., Ltd.

1. Technology Development Goal



1) Background and need for business promotion

(1) Environmental analysis

Export SMEs face complex barriers in overseas sales processes due to translations that do not reflect company-specific contexts, interpretation structures unsuitable for real-time consultation, and the high cost and long duration of regulatory certification document handling. Thus, beyond simple translation functions, a domain-specific AI platform that connects the entire export business cycle based on company data and specialized terminology is needed.

Key barriers	On-site issues	Need for technology development
① Work context·Specialized terminology	Generic AI fails to sufficiently reflect product names, technical specifications, prices, and contract terms per company, leading to a risk of critical information mistranslation during negotiation and contract processes.	Integrate each company's product DB, price list, and glossary with RAG to reflect in real-time in translation, interpretation, and document creation.
② Real-time interpretation performance	The existing 3-stage structure of STT→MT→TTS accumulates a delay of over 2-5 seconds, with limits on background noise, speaker separation, and specialized terminology processing.	Realize interpretation within 2 seconds through E2E streaming interpretation architecture, speaker separation, and specialized terminology injection.
③ Regulatory certification documents	CE·eCTD·FCC·HACCP, etc. have complex national formats and document structures, posing a significant burden of preparation time and outsourcing costs on SMEs.	Automate translation, structuring, and formatting of regulatory documents and support submission document preparation based on expert review (HITL).

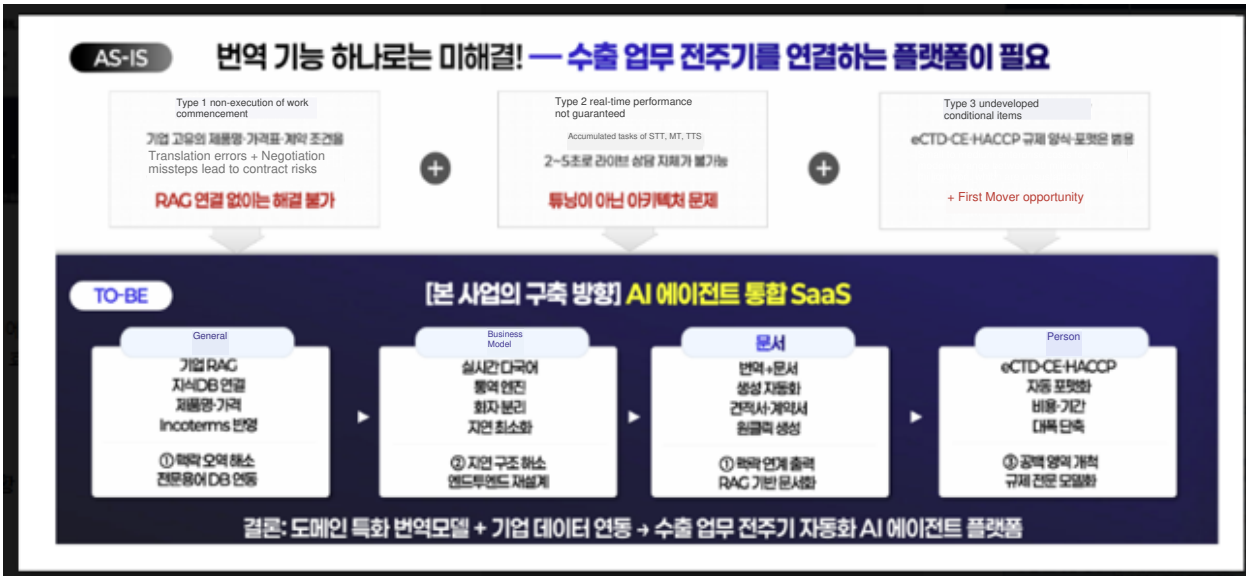
(2) Market growth and development urgency

◦ The global business AI translation market is projected to grow from about $1.8 billion in 2025 to over $7 billion in 2030, with rapidly expanding demand for business domain-specific solutions.

◦ The regulatory certification service market sees substantial costs in document preparation, translation, and formatting, increasing the demand for time and cost savings through document automation. ◦ There is a market opportunity for a SaaS-integrated solution that bridges the gap between general translations and high-cost specialized services for domestic export SMEs.

To connect these market and field demands to actual export performance, improving only the accuracy of translation is not enough. It is necessary to integrate unique company knowledge in real-time translation and interpretation and to link consultation, document creation, and regulatory certification responses into a single workflow. Therefore, this project aims to develop a domain-specific AI platform supporting the full cycle of export business based on company data and specialized terminology.

2) Limitations of existing solutions and the direction of this business establishment (1) Limitations of existing solutions The limitations of existing solutions lie not only in the insufficient performance of individual functions but also in the structural problem where business context, real-time consultation, document, and certification tasks are disconnected. This project will automate the entire export business cycle by connecting it based on corporate RAG and domain-specific models.



(2) Final Development Objective and Three Major Solutions of this Project

Three Major Solutions	Key Development Content	Main Objectives
① Business Translation Model	NLLB/EXAONE-based Domain Learning + Per-enterprise RAG Knowledge DB + Expert Feedback Learning	BLEU 65+ Terminology Consistency 40%↑ Hallucination 70%↓
② E2E Real-time Voice Interpretation	Whisper·InfiniSST-based Streaming Interpretation, Speaker Separation, Real-time Enterprise-specific Terminology Injection	8+ Languages Delay Less than 2 Seconds WER 15%↓
③ Automatic Document Creation· Regulatory Certification	Development of 5 Types of Export Document Agents and eCTD·CE·FCC·HACCP Formatting Modules	TEDS 70%+ Document Creation Time 70%↓ F1 0.8+

(3) Development Direction

This project does not stop at developing the three solutions individually but integrates the domain-specific translation model and enterprise-specific RAG knowledge DB as a common foundation. Through this, the product information and specialist terminology of enterprises are consistently reflected in real-time interpretation and document creation, automating the full operation cycle from overseas sales, buyer consultations, export document creation, to regulatory certification response, on a single AI agent platform.

3) Status of Domestic and Overseas Technology and Competitive Environment

While solutions exist in translation, interpretation, and document processing in domestic and international markets, there is limited case of providing sales, consultation, real-time interpretation, professional translation, and regulatory document automation as a unified SaaS for SMEs. (1) Major Technology Status at Home and Abroad

Function Area	Leading Companies·Services	Main Strengths	Limitations from Project Perspective
Translation	DeepL	Excellent general text translation quality and user-friendliness	Real-time reflection of enterprise-specific terms, prices, and contract conditions, creation of regulatory document structure, linking of business processes is limited
Real-time Interpretation	Google Gemini Live Translate	Low-latency multilingual interpretation feature focused on everyday conversations	Consistency of business-specific terminology, management of multiple speakers, linkages with enterprise documents·RAG is limited
Professional Translation Service	TransPerfect·RWS	Professional personnel and multilingual project operation capability	High-cost, long lead-time structure based on human resources, with limited automation as a SaaS for SMEs and new document creation features
Document Processing·Automation	Hyperscience·UiPath	Capabilities in unstructured document classification·extraction and business automation	Lacks features in export regulation domain, creation of new multilingual documents, and enterprise-specific terminology RAG
Domestic Translation	Flitto	Leading domestic translation platform combining AI and human translation	Limited generation of regulatory documents, regulatory DB, and linkage of the overall export business cycle
Domestic Support·AI	KOTRA Export Voucher, CLOVA·Saltlux·Maindslab, etc.	Possession of translation support or AI documentation·consultation capability	Dependency on human resources or limited proliferation of SaaS specialized for export·trade·regulatory certification and targeting external SMEs

(2) Comparison by Function Area Coverage

Platform	Sales·Consultation	Real-time interpretation	Professional Translation	Regulatory Document Automation	Overall
StunningBiz AI (This Project)	O	O	O	O	Integrated Offering
DeepL	X	X	O	X	Translation-focused
Google	△	O	△	X	General-purpose Focused
TransPerfect·RWS	X	X	△	△	Human Services
Hyperscience·UiPath	X	X	X	△	Document Processing-focused
Flitto·Domestic AI Solutions	△	△	△	X	Partial Support

※ O: Dedicated feature provided / △: Partial support or general-purpose feature response / X: Not supported. This qualitative comparison is based on publicly available materials, and the final submission will indicate the survey reference date and sources.

(3) Conclusion of Competitive Environment Analysis

The comparison shows that general-purpose translation quality, domain expertise in professional translation services, and the automation strength of document processing solutions all exist independently. However, these services have limitations in providing a unified SaaS environment that interlinks enterprise-specific product information and terminology through RAG, extending from sales, consultations, and real-time interpretation to professional translation and regulatory document automation.

Thus, there is a market gap in the domain of integrated AI SaaS for SMEs that supports the full-cycle export operations based on enterprise-specific knowledge.

4) Differentiation·Originality and Technology Development Direction

The uniqueness of this project lies in connecting the workflow from sales→consultation→documentation→certification based on a common foundation of over 1.1 million expert corpora and enterprise-specific RAG knowledge DB, rather than a simple integration of individual functions.

Differentiation Axis	Owned Assets Core Technology	Project Implementation Direction
① Exclusive Expert Data	Over 15 years of accumulated expert parallel corpora 1.1 million+ sentence pairs - Business: 600,000 - Regulatory certification: 500,000	Leveraging domain learning in trade, sales, contracts, and regulatory certification, and building industry-specific terminology DBs to establish data entry barriers that are difficult to replicate in a short period.
② Enterprise Context-Specific RAG	Product DB, Price Lists, Contract Terms, Translation Memory, and Hybrid Search (BM25+Dense Retrieval)	Inject foundation data in real-time for translation, interpretation, and document creation to reduce key errors and hallucinatory issues such as product names, figures, and Incoterms.
③ E2E Real-time Speech Interpretation	Streaming structure based on Whisper and InfiniSST, speaker separation, VAD and noise response	Implement to reduce the accumulated delay of the existing three-stage Cascade and to achieve interpretation within 2 seconds during multilateral buyer consultations and exhibitions. Develop a
④ Export Business Integrated SaaS	5 types of export document agents, eCTD, CE, FCC, HACCP formatting, Web, Mobile, API	service that can link corporate data within at least three days and expand to shopping malls, CRM, ERP, and meeting platforms.

(1) Direction for Verification and Commercialization Connection

Through field testing by three demanding companies, we verify not only translation accuracy but also the potential for reducing work time, improving document quality, and enhancing export response speed and order acceptance rates. ◦ AI outputs are provided as drafts, and regulatory certification documents are subject to mandatory Human-in-the-loop (HITL) review procedures to manage legal and business risks.

◦ Combining PCN's SaaS, API, and cloud operational capabilities with StunningBox's AI model, data, and regulatory document capabilities to simultaneously demonstrate and commercialize within 24 months.

This project aims to resolve the gap between generic translation AI and high-cost specialized services, based on expert data and enterprise-specific RAG, and develop and demonstrate a B2B domain specialized AI SaaS that integrates the entire export business cycle.

2. Research and Development Method

1) Data Acquisition Method

This dataset is constructed based on 15 years of accumulated real transaction parallel data obtained through an official data utilization contract between the joint R&D entity, StunningBox Inc., and the 'Korean Translation Institute'. It consists of outputs from actual professional translation projects carried out for various SMEs. The raw data provider's security regulations and the Personal Information Protection Act are strictly adhered to, with de-identification and anonymization completed. After that, StunningBox's proprietary data refinement technology was used to perfectly reprocess and assetize the dataset for AI training. The data utilization contract for this project's execution and the de-identification measures have been reviewed and completed.

Data Acquisition Method — StunningBiz AI Translation and Interpretation Platform 1. StunningBox's own business and regulatory parallel corpus [Core Differentiation - Entry Barriers] Business collaboration and negotiation dialogues, product descriptions, catalogs, proposals, clinical, regulatory, and certification documents, contracts, trade documents Korean-English parallel 300,000 sentence pairs • cCTD-KC-CE-FCC 500,000 sentence pairs approximately 100,000 sentence pairs Total: Over 1.1 million sentence pairs possessed (15 years of accumulated expert dataset) — Exclusive data that competitors cannot easily replicate in a short time 2. Field data collection for three demanding companies - Industry-specific empirical data acquisition Sunlight Lucorn [Automotive Parts] HB [Footwear Manufacturing] Grey Hear [K-Beauty] English German Japanese English Chinese Vietnamese English Chinese Japanese Technical documents Negotiation dialogues CE FDA 43 documents BOM, OEM contracts, etc. catalogs K-Beauty marketing content Exhibition scripts raw material regulatory agency documents 3 industries × Actual collected astronomical data — Foundation data for building industry-specific RAG simulators — Basis for expanding into all industries Data refining De-identification RAG Vector DB embedding AI model training Fine-tuning Internalization Joint construction of industry-specific terminology DBs (Terminology DB) tailored to demanding companies, design of translation interpretation quality feedback loop, and Active Learning application Additional collection of public data: KOTRA overseas market news, Korea Customs Service HS code DB, International Certification DB (FDA CDER, EMA EudraLex, KS/ISO) Crawling and refining multilingual parallel data of product descriptions from overseas e-commerce sites (Amazon·Alibaba·Rakuten), and collecting integrated data linking domestic e-commerce sites (1) Data Quality Management Plan a. Core Data Validation Pipeline Build learning and verification datasets based on the scale of learning data, expert review and effect verification Data Refining Over 1.1 million 15-year accumulated expert dataset Negotiation and Collaboration Dialogues 300,000 catalogs and proposals 200,000 Collected Data Regulatory and Certification Documents 500,000 Contracts and Legal 100,000 KOTRA-Korea Customs Service HS Code International Certification DB Selected Parser, De-identification Blocking, Labeling, Vector DB, Governance (IAA 0.85+) RAG Vector • Feature/Label Dataset DB construction / Proposal Evaluation AI Model Training Fine-tuning RLHF Domain Fine-tuning Quality Evaluation and Verification Feedback Loop Labeling Guidelines Learning Verification Expert Review and Effect Verification Expert Evaluation (MQM - Field) Translation Quality Evaluation : BLEU Measurement Machine Translation Performance (WER) Negotiation and Legal Verification Regulatory Document Compliance Effect Measurement (A/B Testing 3 times) Artificial Intelligence Market Daily 2 AI2+ 3,500 Linkage Consistency Test KOLAS BLEU-WER Accredited Verification Feedback Loop Expert Review & RLHF Training: Monthly Client Provision



Stage	Execution Contents
① Collection	Collection of Source Data by Demand Company ② Refinement MinHash-based duplicate removal, Rule-based cleansing (ungrammatical errors, typos filtering)
③ Verification Sample	cross-review by professional trade and translation personnel, anomaly detection
95%+ Accuracy, Duplication Rate below 5%, and Error Rate below 3%. Only pass data (Gold Dataset) ④ Learning Reflection	Incorporate into Fine-tuning

(2) Legal Responsibility and Expert Review of Automated Regulatory Documents
(HITL) HITL Review Procedure AI Draft Generation → Expert Review → Approval/ Rejection → Final Issue (issuance of the final version is not possible without review)
Accountability, Notification, Update System

Item	Content
Legal Responsibility Structure	Clearly notify that AI outputs are at the draft status, and the final submission responsibility lies with the user (demand company). Insert the domain expert translation reviewer step as a mandatory workflow. Check updates to country-specific export regulation and customs guidelines at least once a quarter and update rule sets. Post notification phrases within the service and prepare exemption clauses in the contract.
Expert Review Procedure	
Guideline Update by Country	
User Notification and Disclaimer Structure	

(3) Data Security Four-layer Structure

Stage	Execution Contents
① Encryption at Rest	Encrypt DB/File Stored Data with AES, etc.
② Encryption In Transit	TLS-based encryption of transmission sections
③ Access Authority Control	Role-Based Access Control (RBAC)
④ Audit Log Record	Track the history of access and changes meticulously

2) AI Model Development Method

(1) Business Hybrid Translation Language Model (3-Stage Training Pipeline)

Stage	Method	Core Content
Stage 1 Domain Fine-tuning	Transfer Learning with Business Corpus	Internalization of domain language patterns based on trade, sales, and contract business parallel corpora (500,000 sentence pairs). Industry-specific specialized learning for three demand company industries.
Stage 2 RAG Integration	Enterprise-specific Knowledge DB Real-time Search Augmentation	Store unique product names, prices, and conditions in vector DB. During translation/interpretation, inject real-time context via Hybrid Search (BM25+Dense Retrieval). Aim to reduce hallucination by 70%.
Stage 3 RLHF	Expert Feedback Preference Learning	Reinforcement learning based on feedback from translation experts and sales practitioners. Optimization of business negotiation style, honorific speech, and cultural nuances.

※ Of the 600,000 business parallel corpus sentence pairs possessed, a training subset of 500,000 sentence pairs is utilized for the first-stage fine-tuning.



(2) E2E Real-time Streaming Voice Interpretation Model. [Technical Architecture]
InfiniSST + Whisper Large-v3 + LLM-based E2E Streaming Interpretation

Technical Element	Implementation Method and Content
Chunked Causal Streaming ASR	Whisper Large-v3-based Chunk Level Causal Encoder. Real-time processing during speech. Business jargon and product names specialized fine-tuning. Background noise filtering with VAD (Voice Activity Detection). WER target below 15%.
E2E Simultaneous Interpretation Core Model	InfiniSST Framework: Wav2Vec2.0 Encoder + Adapter + Qwen2.5-7B LLM Decoder. Handles unlimited voice streams with sliding KV-Cache management. Apply AlignAtt simultaneous interpretation policy. Target delay: StreamLAAL below 2 seconds.
Business Context RAG Real-time Integration	Real-time referencing of corporate RAG knowledge DB during interpretation (initial prompt injection). Prevents mistranslation of product names, model numbers, prices, and Incoterms. Domain term beam search weight applications.
Multilingual Support & Speaker Separation	Korean <-> English, Chinese, Japanese, Spanish, German, French, Vietnamese, Arabic (8+ languages). Automatic speaker differentiation with Speaker Diarization. Automatic language detection per speaker.
Platform Integration	Native plugins for Zoom, MS Teams, Google Meet. Mobile app (iOS, Android). Low-latency streaming API for exhibitions and field use. Supports offline on-premises deployment (sensitive information security).

(3) Export Document Automation Agent - 5 Core Operations

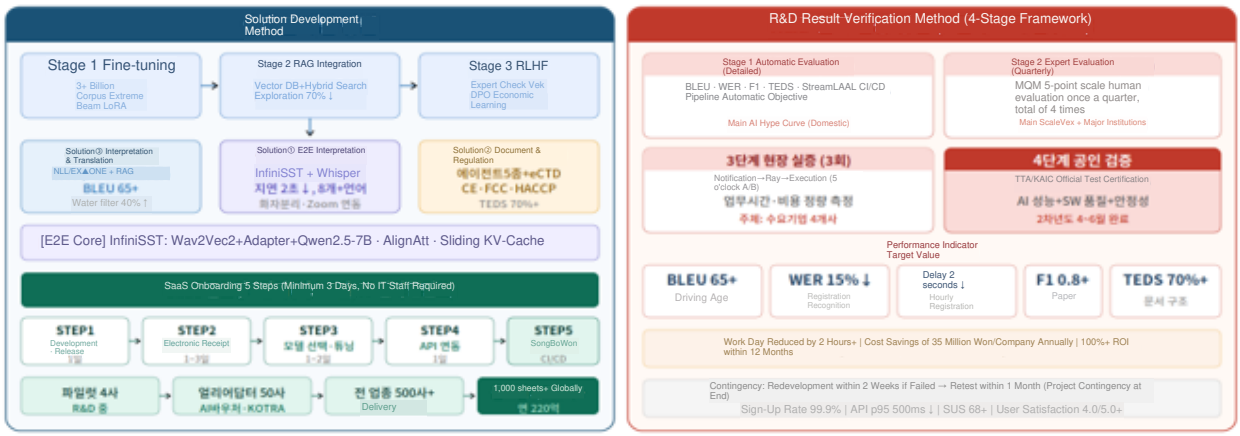
Document Type	AI Agent Processing Content and Enterprise Utilization
Multilingual Product Pages & Catalog	Korean product description -> Multilingual (English, Chinese, Japanese, Spanish, French) shopping mall optimized phrase automatic generation. SEO keyword and cultural localization. [Health Functional Foods, Cosmetics] Cafe24, Amazon API integration.
Quotation & Quote	Reference unit price list with RAG -> Reflect trade standards such as Incoterms, payment terms, packaging units. [Automotive Parts, Footwear] Automatic multilingual generation for OEM delivery quotations.
Auto Response for Overseas Inquiries	CRM, shopping mall API: Foreign language inquiry -> Intent classification -> RAG search -> Automatic multilingual response. [Food] Automation of frequently asked questions related to ingredients, allergies, certifications.
Multilingual Contract Template	Supply contracts, NDA, MOU templates + enterprise terms injection -> Automatic multilingual draft creation. [Automotive Parts] Global OEM supply contract multilingualization.
Exhibition, Seminar Real-time Interpretation Support	Voice interpretation + multilingual subtitles + Q&A chat translation integration. Zoom, Teams plugins. [Footwear, Cosmetics] Real-time consultation support at overseas exhibitions (MIDO, Cosmoprof).

(4) Regulatory Certification Document Formatting Module [StunningBox Exclusive Differentiation]

Certification Type	Target Enterprise	Automation Content
eCTD (Electronic Common Technical Document) FDA, EMA, PMDA, NMP A	Functional Health Foods (FDA Registration) Cosmetics (NMPA China Registration)	ICH M4 5-module automatic formatting, Module 1 country-specific automatic generation, XML Backbone, PDF/A automatic conversion.
CE Marking (EU Conformity Declaration)	Automotive Parts, Footwear Cosmetics (EU Export)	EC Declaration of Conformity automatic formatting, Technical Documentation translation and structuring.
FCC Certification (U.S. Radio Certification)	Electronics with electronic or wireless communication function and automotive parts	FCC Form 731 automatic formatting, technical document English translation and structuring.
HACCP, Food Export Documents	Food Manufacturers, Functional Health Foods	Automatic multilingual generation of ingredient tables, allergy labels, origin certificates, automatic application of country-specific labeling standards.

3) Solution Development and Distribution Method to Target Enterprises

The AI Interpretation and Translation Solution of this project has already completed PoC in the global e-commerce field of genuine grip, proving commercialization viability. Multilingual product page automatic generation and regulatory document translation are operated as a real service, and in this consortium, this verified AI core is mounted on the PCN SaaS platform, and expanded to all industries.



(1) Solution Development — 3-Stage Training Pipeline → Output of 3 Main Solutions

Classification	Core Technology	Development Content	Target Performance
Solution①	Business Translation Model	3-Stage Training Based on NLLB/EXAONE (Fine-tuning → RAG → RLHF). Utilizing 1.1 Million Corpus of Study Box. 3 Types of Industry-specific LoRA Adapter. Real-time Context Injection for Companies' RAG	BLEU 65+, Terminology Consistency 40%↑, Hallucinations 70%↓
Solution②	E2E Real-Time Speech Interpretation	InfiniSST: Wav2Vec2+Adapter+Qwen2.5-7B LLM. AlignAtt Simultaneous Interpretation Policy. Unlimited Stream with Sliding KV-Cache. Bidirectional in 8+ Languages. Speaker Separation, Zoom/Teams Integration	StreamLAAL 2s↓, WER 15%↓, DER 15%↓
Solution③	Document Auto-Generation · Regulatory Certification	Five Types of Agents (Catalog · Quotation · Auto Response · Contract · Expo Interpretation). Regulatory Certification Formatting (eCTD · CE · FCC · HACCP). 500,000 Corpus of Study Box for Proprietary Differentiation	TEDS 70%+, Document Generation Time 70%↓, F1 0.8+

(2) Distribution Method for Demand Companies

Stage	Required	Execution Contents	Deliverables	Value from Company's Perspective
1	1 Day	SaaS Signup → Plan Selection (Starter/Business/Enterprise) → Payment → Automatic Cloud Tenant Creation	Enterprise-Specific SaaS Instance	No Separate Server/IT Staff Needed. Immediate Use Upon Card Payment
2	1~3일	자사 데이터 업로드(제품DB·용어집·가격표) → 자동 파싱·정제 → RAG 벡터DB 임베딩	기업 전용 RAG 지식DB	기존 용어·제품명 즉시 반영. 범용 AI 대비 정확도 40% ↑
3	1~2일	업종별 사전학습 모델 선택(제조/도소매/식품/IT) → 경량 LoRA 파인튜닝 → A/B 테스트	기업 맞춤 번역 모델	업종 특화 모델로 전문 용어 정확도 극대화
4	1일	API 키 발급 → 쇼핑몰(Cafe24·Amazon)·CRM·ERP 연동 → 테스트 실행	API 연동 완료, 실서비스 시작	기존 업무 시스템에 자연 통합. 추가 개발 불필요
5	Continuous	Usage Logs → Quality Monitoring Dashboard → Monthly CI/CD Automatic Retraining → Automatic Glossary Updates	Monthly Quality Reports, Model Autonomous Improvement	An Auto-Learning System That Becomes More Accurate with Use

(3) Core Requirements and Reflections of Demand Companies (3-1) Key Functions by Requirement a. Structural Bottleneck in the Industry at Large

Item	numerical value	meaning
국내 수출기업 수	약9.3만 개사(2024, 관세청)	3개 수요기업이 겪는 문제는 유사 업종·규모 기업 전반이 공통으로 겪는 구조적 문제
국내 수출문서 외주 시장 규모	연간 약3.8조 원	수작업·외주 의존 구조가 산업 전체에 고착된 고비용 문제
글로벌 언어서비스 시장	539억 달러	언어 서비스 자체가 대규모 성장 시장이며 자동화 수요가 구조적으로 존재
글로벌 컴플라이언스 소프트웨어 시장	362억 달러(CAGR 12.67%)	규제 대응 자동화가 전체 시장 성장률의1.5배 속도로 고성장— 시장이 자동화 솔루션을 요구하는 방향으로 이동 중
certification/document response costs per country	hundreds to tens of millions of KRW+ several months lead time (Repeat occurrence by industry·country)	The cost and time of exports to multiple countries accumulates repeatedly by the number of countries × the number of products

Source: Korean Customs Service (2024), KTR Export Certification Support Center (2024), Element Korea Overseas Certification Cost Guide (2024), KOTRA Export Voucher Survey (2024), Mordor Intelligence (2025) summary based

b. Connection of Demand Companies' Respective AS-IS to Industry Statistics

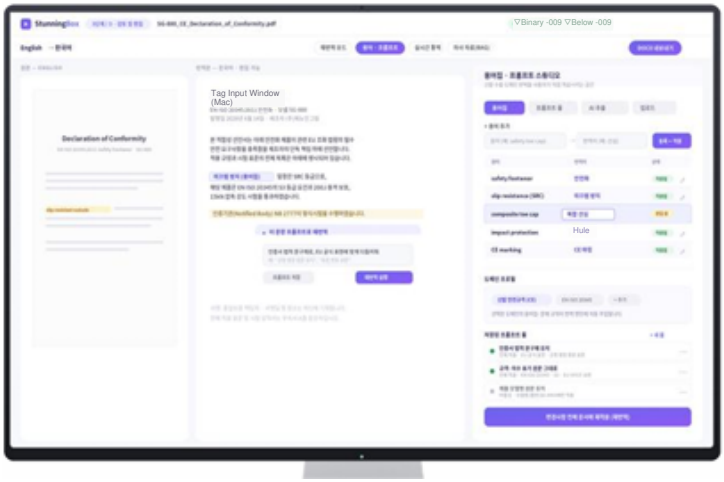
Demand Companies	Individual AS-IS (Current)	Connection to Evidence Across the Industry
Sunlight Illucom (Automotive Parts)	Multilingual product manual translation/ creation 72 hours	Even automotive parts and other manufacturing export companies repeatedly incur structural costs for country-specific certification and manual responses—Even large corporations (such as Kia Motors) experience similar bottlenecks by concurrently using in-house translation systems and external agencies. Consumer goods export sectors like K-Beauty incur excessively high costs related to country-specific regulation and consultation compliance relative to sales volume—The high growth of the language service market (53.9 billion USD) itself evidences this demand Special sectors such as highly regulated shoe industries experience hundreds to thousands of KRW per country and several months of lead time—A typical case of a 3.8 trillion KRW outsourcing market
Gray Whale (K-Beauty)	Global buyer email and real-time consultation response average 24 hours	
Genuine Grip (Special Footwear)	Country-specific certification/ regulatory document analysis/ draft preparation 48 hours	

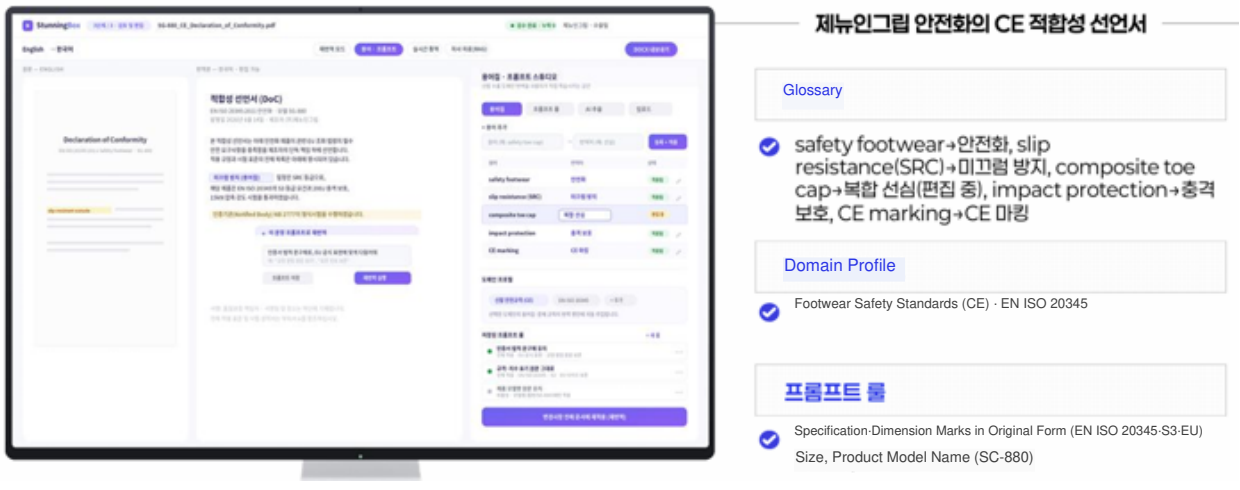
Demand Companies	Requirements	Core Technologies for Response	Performance Indicators
햇빛일루콤	다국어 카탈로그 용어 일관성 요구	온톨로지 기반 용어 제약 디코딩	전문용어 F1
그레이웨일	실시간 바이어 상담 대응 지연 최소화	StreamLAAL 기반 실시간 통역 파이프라인	처리속도(2초 이내)
제뉴인그립	해외 규제문서 표/형식 보존 요구	TEDS 기반 구조보존 문서생성 에이전트	문서구조 보존율(TEDS)

(3-2) Practical Use Plan and Advancement Plan

a. Practical use: Web Dashboard Construction

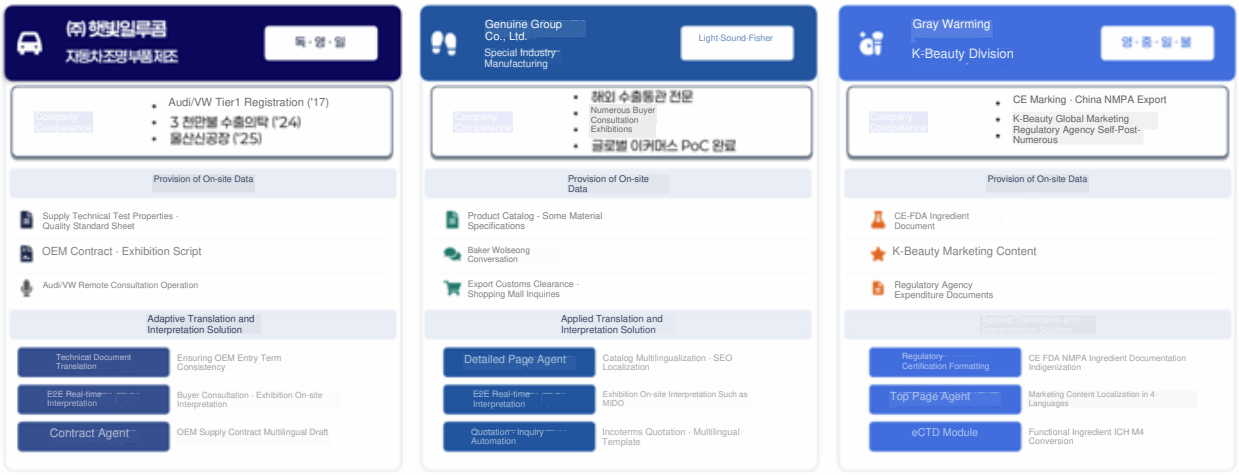
- API integration, choice of company-specific application method during linkage to demand company internal business systems





-Onboarding: Minimum 3 days per standard data baseline, typically real-use launch within 3 to 7 business days post data upload - Advancement: Quarterly model update, automatic glossary reflection, history-based consultation - Continuous Learning Pipeline Operation

b. Verification Plan for Demand Companies



(3-3) Field Application Effects for Demand Enterprises (AS-IS, TO-BE)

Demand Companies	AS-IS	TO-BE (After Introduction, Interim Goal)	Key Improvement Indicators*
햇빛일루콤	다국어 제품 매뉴얼 번역/생성 −72시간	AI 적용 후 24시간 이내 (감수 포함)	매뉴얼 번역시간 50% 이상 단축
그레이웨일	글로벌 바이어 이메일·실시간 상담 대응 평균 − 24시간	AI 적용 후 30분 이내	상담 응대 리드타임 97% 대폭 단축
제뉴인그립	국가별 인증·규제문서 분석/초안 작성 − 48시간	AI 적용 후 24시간 이내	규제문서 처리시간 70% 이상 단축

*Methodology for Calculating Key Improvement Indicators (Same as Announcement Performance Indicators Calculation Methodology) ① Comprehensive Survey of Demand Enterprises→ ② Exclude Outliers Based on Quartile Standard→ ③ Calculate Average (Determine AS-IS) → ④ Set Improvement Rate Goal (TO-BE) Apply the same four-step procedure to all three demand enterprises to ensure the objectivity of AS-IS/TO-BE figures.

(4) Scheme for Securing Performance Verification System

Apply four-step verification (Automatic→Expert→On-site→Certified). Continuously measure 5 KPI including BLEU 65+, WER 15%, and if underperform, redevelopment within 2 weeks. Verified in demonstration with three demand enterprises that work time reduced by 2+ hours per day, and cost savings of 35 million KRW per enterprise per year. ※ Key: Development→Dissemination→Verification are conducted simultaneously in an Agile manner and on-site feedback is immediately reflected in RLHF, implementing SaaS that becomes more accurate with CI/CD automatic retraining as it is used more frequently.

1) Performance Indicators and Targets

Stage / Performance Indicator Name	First Year	Second Year	Third Year	Total	Weight (%)
Patent Application	1	1	1	3	25
Patent Registration	-	-	1	1	25
Expert Report	2	2	2	4	25
SW Registration	-	-	1	1	25
Total	3	3	5	9	100

※ The one patent registration in the third year is for the patent applications (3)

2) Performance Indicators and Measurement Methods

(1) Performance Indicators of Outputs

Classification	Evaluation Items (Key Performance)	Unit	Proportion in total Items	World-Class Level (Holding Country/ Institution)	Domestic Level Before R&D	R&D Target Value			Basis for Setting Targets
				Performance Level	Performance Level	First Year	Second Year	Third Year	
Model Performance	Business Domain Translation Accuracy (BLEU)	score	20	USA/Google Gemini 3.5 Live (70 languages real-time), Germany/DeepL However, Business Domain·B2B Technical Terms Not Supported	Centered around general-purpose translation services like Papago and Flitto, lacking performance measures and domain-specific solutions.	-	60 or more	65 or more	Application of SacreBLEU, the standard metric for WMT.
	Interpretation delay time (StreamLAAL)	sec	15	Delay of 2.5~3 seconds or more, such as by the USA/ Google S2ST (with no Korean support)	Centered around general-purpose translation services like Papago and Flitto, lacking performance measures and domain-specific solutions.	-	2.5 seconds or less	2 seconds or less	Aiming to achieve world-class standards, applying ACL StreamAtt's StreamLAAL evaluation method.
	Speech Recognition Error Rate (WER/CER)	%	15	For general domain by USA/OpenAI Whisper Large-v3, WER is 10~12%	Centered around general-purpose translation services like Papago and Flitto, lacking performance measures and domain-specific solutions.	-	20% or less	15% or less	Adoption of the Whisper's Common Voice·FLEURS WER/CER evaluation method.
	Document Structure Preservation Rate (TEDS)	%	10	–	–	-	60% or more	70% or more	Application of the TEDS standard metric of PubTabNet.
	Term Extraction (F1)	score	10	Based on common glossary like UK/ RWS Holdings SDL Trados, F1 level is 0.6~0.7	Remains at the level of static registration features like the Papago Plus glossary function, lacking automatic extraction and real-time search augmentation systems.	-	0.7 or more	0.8 or more	Precision and Recall evaluation based on approved Gold Standard glossaries from demand companies.

Classification		Evaluation Items (Key Performance)	Unit	Proportion in total Items	World-Class Level (Holding Country/ Institution)	Domestic Level Before R&D	R&D Target Value			Basis for Setting Targets
					Performance Level	Performance Level	First Year	Second Year	Third Year	
현장 적용 (솔루션 도입 전·후 효과성)	Haeppit Illucom	Product manual translation	Time	5	-	72	-	60	54	Reduced the average working time from 72 hours to 18 hours.
		Understanding and translating the context of entire email threads	Time	5	-	4	-	1,5	2	Reduced daily work time for trade personnel from 4 hours to 2 hours
		Translation of regulatory certification documents	ten thousand won	5	-	1,000	-	600	500	Reduced business costs by 50%
	Gray Whale	Global shopping mall page	Cases	5	-	-	-	2	3	Due to lack of domain-specific global translation, created pages in three languages (English, Chinese, Japanese)
		Real-time buyer consultation interpretation	Sec	5	-	6	-	2.5	2	Achieved interpretation delay time of 2 seconds
		Understanding and translating the context of entire email threads	Time	5	-	4	-	1,5	2	Reduced daily work time for trade personnel from 4 hours to 2 hours
	Genuine Grips	Translation of regulatory certification documents	ten thousand won	5	-	300	-	180	150	Reduced business costs by 50%
		Product manual translation	Time	5	-	72	-	60	54	Reduced the average working time from 72 hours to 18 hours.
		Understanding and translating the context of entire email threads	Time	5	-	4	-	1,5	2	Reduced daily work time for trade personnel from 4 hours to 2 hours

※ The domestic level before the field application of R&D corresponds to the level before the solution adoption by the demand companies.

(2) Evaluation Methods and Environment

Evaluation Items (Performance Indicators)	Evaluation Environment (Measurement Method)	Evaluation Method
Business Domain Translation Accuracy (BLEU)	• **평가내용** : FLORES-200 및 규제문서 Test Set(1,000문장)을 전문가 번역문과 비교 • **평가절차** : AI 번역 결과와 정답 번역문을 SacreBLEU로 비교 평가 • **산출식** : SacreBLEU(Test, Reference) • **변수설명** : Test=AI 번역문, Reference=전문가 정답 번역문	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Interpretation Delay Time (StreamLAAL)	• **평가내용** : 한국어↔영어 실시간 통역 100문장 수행 • **평가절차** : 발화 시점부터 번역 출력까지의 평균 지연시간 측정 • **산출식** : Σ(g(t)−t) / N • **변수설명** : g(t)=AI 번역 출력시점, t=사용자 발화시점, N=전체 문장수	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Speech Recognition Error Rate (WER/CER)	• **평가내용** : 회의 음성 10시간과 정답 스크립트 비교 • **평가절차** : 치환·삭제·삽입 오류를 계산하여 오류율 산출 • **산출식** : (S+D+I) / N ×100 • **변수설명** : S=치환(Substitution), D=삭제(Deletion), I=삽입(Insertion), N=전체 단어수	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Document Structure Preservation Rate (TEDS)	• **평가내용** : PDF 100건을 HTML로 변환 후 원본과 비교 • **평가절차** : 문서 구조(Tree) 유사도를 자동 계산 • **산출식** : TEDS(Tree Edit Distance Similarity) • **변수설명** : 원본 HTML 구조와 변환 결과의 구조 유사도(0~1)	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Term Extraction (F1)	• **평가내용** : 규제문서 Test Set(3,000문장) 분류 평가 • **평가절차** : Precision과 Recall을 산출 후 F1 계산 • **산출식** : 2×P×R / (P+R) • **변수설명** : P=Precision(정밀도), R=Recall(재현율)	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Product Manual Translation	• **정의** : AI를 활용한 제품 매뉴얼 번역 업무의 작업시간 단축 효과 • **평가내용** : 제품 매뉴얼 5종(총 10건)을 대상으로 AI 도입 전·후 번역시간 비교 • **평가절차** : 동일 문서를 기존 방식과 AI 방식으로 번역하여 평균 작업시간 측정 • **산출식** : 평균 단축시간 = (AI 도입 전 작업시간 − AI 도입 후 작업시간) / 시료수	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Understanding and Translating Email Thread Context	• **정의** : 이메일 스레드의 맥락 이해 및 번역 업무시간 단축 효과 • **평가내용** : 바이어 이메일 스레드 20건을 대상으로 AI 도입 전·후 업무시간 비교 • **평가절차** : 동일 담당자가 기존 방식과 AI 방식을 적용하여 일평균 작업시간 측정 • **산출식** : 일평균 단축시간 = AI 도입 전 업무시간 − AI 도입 후 업무시간	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Translation of Regulatory Certification Documents	• **정의** : 규제·인증 문서 번역 비용 절감 효과 • **평가내용** : 규제·인증 문서 10건을 대상으로 기존 외주방식과 AI 적용 비용 비교 • **평가절차** : 동일 문서의 번역비용을 AI 도입 전·후로 산정하여 비교 • **산출식** : 비용 절감률 = (도입 전 비용 − 도입 후 비용) / 도입 전 비용 × 100	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Creation of Global Shopping Mall Pages	• **정의** : 글로벌 쇼핑몰 다국어 페이지 자동 생성 지원 범위 • **평가내용** : 대표 제품 10종을 대상으로 다국어 상품페이지 생성 • **평가절차** : 언어별 생성 결과를 검수하여 지원 가능한 언어 수 확인 • **산출식** : 지원 언어 수 = 품질기준을 만족하는 생성 언어 수	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Real-time Buyer Consultation Interpretation	• **정의** : 바이어 상담 시 실시간 통역 응답 지연시간 • **평가내용** : 한·영 상담 발화 100건을 대상으로 평균 응답시간 측정 • **평가절차** : 발화 종료 시점부터 번역 출력 시점까지의 시간을 자동 측정 • **산출식** : 평균 지연시간 = Σ(번역 출력시점 − 발화 종료시점) / 발화 수	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Understanding and Translating Email Thread Context	• **정의** : 이메일 스레드의 맥락 이해 및 번역 업무시간 단축 효과 • **평가내용** : 바이어 이메일 스레드 20건을 대상으로 AI 도입 전·후 업무시간 비교 • **평가절차** : 동일 담당자가 기존 방식과 AI 방식을 적용하여 일평균 작업시간 측정 • **산출식** : 일평균 단축시간 = AI 도입 전 업무시간 − AI 도입 후 업무시간	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)
Translation of Regulatory Certification Documents	• **정의** : 규제·인증 문서 번역 비용 절감 효과 • **평가내용** : 규제·인증 문서 10건을 대상으로 기존 외주방식과 AI 적용 비용 비교 • **평가절차** : 동일 문서의 번역비용을 AI 도입 전·후로 산정하여 비교 • **산출식** : 비용 절감률 = (도입 전 비용 − 도입 후 비용) / 도입 전 비용 × 100	Second Year: Test Report (Demand-side Evaluation) Third Year: Official Test Report (Demand-side evaluation and official testing agency witness test)

(5) Quantitative Comparison Indicators— Analysis of Numerical Gaps In addition to qualitative coverage comparison, a quantitative comparison based on actual disclosed figures is presented. Figures are based on each company's public data and announcements, and the platform's target figures remain consistent with the previously presented performance indicators comparison table (items 5, 6, 12, 20 combined).

Comparison Items	Unit	Competitors/General Standards	numerical value	Platform Target
Translation Quality (Expert Preference)	Multiple	GPT-4 (vs DeepL)	1.7 times lower preference compared to DeepL (DeepL's own blind test)	Aim to achieve equivalent or superior levels with domain-specific fine-tuning (to be verified with internal A/B benchmark)
Translation Post-Editing Requirement	Multiple	GPT-4 (vs DeepL)	Requires 3 times more editing compared to DeepL (based on DeepL's own announcement)	Aim to minimize editing needs with specialized terminology consistency metric (over 85% term matching rate)
Real-time Interpretation Delay	Sec	Transync AI·Tiro etc. (General-purpose Conversation)	Less than 0.5 seconds	Within 2 seconds (includes conditions for consistency of technical terms + RAG search + document linkage)
Document Preparation Lead Time (Human-based)	Duration	TransPerfect·RWS etc.	Weekly Unit	Target for Demand Companies: Within 2 hours (up to 97% reduction)
Translation Rate (Domestic Professional Agency)	KRW/ Page	Domestic Professional Translation Agency (based on 550 characters)	Typically 20,000 to 50,000 KRW, up to 150% surcharge for specialized fields (contracts, etc.)	Significant reduction through AI automation (exact savings rate to be confirmed after internal cost estimation)
AI Solution Construction Costs	KRW	Hyperscience etc. IDP Solutions	Hundreds of millions to billions of KRW	No initial construction costs due to SaaS subscription model
Regulatory Certification Document Response Costs (Per Country)	KRW·Duration	Average for Domestic Export Companies (StunningBiz AI Survey Data)	Tens to hundreds of millions of KRW + several months lead time	Goal to significantly reduce through automation (verified through demand company demonstration)

※ The 'Real-time Interpretation Delay' item clearly states that a competitor leads in sheer speed— however, the competitor's product is solely for casual conversation and lacks features for consistency of technical terms, RAG search, and document linkage. It's important to specify in the notes that this platform aims for under 2 seconds, despite including these features. Translation rates, construction costs, and regulatory response costs are areas currently lacking precise reduction figures, pending internal cost assessments and verification through demand company demonstration.

3. Preceding Research and Development

1) Preceding R&D History

(1) Leading Organization PCN Co., LTD's Preceding Patents and Technologies

Task Name/Patent Name	Core Achievements/Content	Organization·Duration	Remarks
Ontology-based Search Service System and Method	Implementation of a semantic-based search engine. Technology to improve search accuracy through term relationship inference	PCN Co., LTD March 2019 Registered No. 10-1958811	Patent Registered
Intelligent Search System and Method Based on RDF Triple Data	Structured data-based intelligent search. Preceding technology for RAG vector search	PCN Co., LTD April 2019 Registered No. 10-1972127	Patent Registered
Open API-based Content Service Interface Provision System and Method (OASIS)	API auto-generation platform. Secured architecture for multiple API integration and linkage	PCN Co., LTD August 2013 Registered No. 10-1296321	Patent Registered GS Certification
Oz-i AI Platform - STT-based Automatic Meeting Summary - Multilingual Translation Service	Integrated platform applicable to industrial sites, metaverse, and robots with AI. Completed implementation of STT speech recognition + multilingual translation + sLLM. Applied to the aviation industrial complex metaverse project	PCN Co., LTD 2024~Present	Commercial Platform Multilingual Translation PoC
Supreme Court Criminal Electronic Litigation System Construction (110 billion scale)	Verified capabilities in large-scale platform construction and operation. Capability in public SI business execution	PCN Co., LTD 2023 Contract Awarded	Public SI Large Reference

(2) Collaborating Organization Stunning Box Co., LTD and Subcontractor's Preceding Technologies

Task Name/Patent Name	Core Achievements	Organization·Duration	Remarks
RAG (Search-Augmented Generation) Solution Utilizing Generative AI	Patent application and progress for <RAG (Search-Augmented Generation) Solution Utilizing Generative AI> by Kumoh National Institute of Technology Industry-University Cooperation Group (Inventor: Professor Seong-ryeol Kim)	Kumoh National Institute of Technology -> Stunning Box Co., LTD	Patent Application 10-2024-0147669
Clinical·Regulatory Professional Translation Multi-Agent Prototype	8-types Multi-Agent Translation Pipeline PoC, regulatory document RAG verification, document structure preservation rendering	Stunning Box Co., LTD 2024~Present	B2B PoC Completed

2) Plan for Utilization of Preceding R&D Results in the Execution of This Project

Organization	Preceding R&D Outcomes	Specific Utilization Method for This Project	Period·Cost Reduction Through Utilization
PCN	Oz-i AI Platform (STT·Multilingual Translation·sLLM)	Directly utilized as the base infrastructure of the SaaS platform for this project. Combined Oz-i's STT engine with Whisper Large-v3 as a preprocessing module for E2E voice interpretation. Directly leveraging sLLM serving architecture (vLLM, TensorRT-LLM) for deployment of translation and interpretation models.	Approximately 6 months reduction compared to new platform development. Approx. 50 million KRW reduction in cloud infrastructure construction costs
PCN	Ontology Search Patent (Semantic - based Search)	Applied to the core design of the RAG vector search engine. Implemented the ontology-based semantic similarity algorithm on ChromaDB/Pinecone with a hybrid search layer (BM25+Dense Retrieval). Directly contributes to improving semantic search accuracy for corporate terms and product names.	Approx. 3 months reduction in RAG search engine design. No separate search algorithm research needed
PCN	OASIS API Platform (GS Certification)	Reused RESTful API gateway architecture for integration with demand companies, shopping malls, and CRMs. Applied OASIS's API auto-generation, documentation, and version management functions to the API B2B2B service of this project. Basis for API integration designs of Cafe24, Amazon, Naver Shopping Mall.	Approx. 4 months reduction in API integration module development. Approx. 30 million KRW reduction in API standardization and documentation costs
PCN	Cloud SaaS Operational Capability (25 Years Record)	Directly applied know-how in designing and operating multitenant SaaS architecture. Embedded operating systems such as Kubernetes auto-scaling, CI/CD pipeline, monitoring (Prometheus+Grafana) into this project's SaaS. Technical basis for ensuring 99.9% SLA availability.	Approx. 4 months reduction in SaaS operational system setup. Approx. 20 million KRW savings in stability verification costs
StunningBox	RAG Solution Patent	Directly incorporated into the core patent technology of this project's RAG architecture. Applied on-device secure search technology to protect corporate confidential data (price lists, contract terms). Provides theoretical basis for the Evidence Grounding method to prevent LLM hallucination.	RAG core design approximately 3 months reduction. Security search separately developed not required
StunningBox	Multi-Agent Translation PoC (Document Structure Preservation)	Directly utilized as a basis for the development of '5 export document automation agents' in this project from the 8 types of Multi-Agent pipelines. Applied the eCTD document structure preservation technology verified in the PoC to the regulatory certification formatting module. Reflected in the strategy to achieve professional terminology extraction F1 0.8+ with the RAG validation results of clinical terms	Agent Prototype approximately 4 months reduced. Regulatory document structure analysis costs reduced by approximately 40 million KRW
StunningBox	1.1 million+ professional translation corpus (Accumulated over 15 years)	Immediately deployed as core data for AI model training in this project. 50 million sentence pairs of regulatory corpus → trained regulatory certification formatting module. 60 million sentence pairs of business corpus → fine-tuning of translation models + RAG glossary. Possibility to commence model training from the first year without separate data collection and cleansing	Data collection and cleansing approximately 8 months reduced. Data construction costs reduced by approximately 200 million KRW (maximum differentiation)

4. R&D and Commercialization Capability

1) Plan and Capacity for Team Formation

(1) Structure and Role Allocation of the Consortium

a. Why this Consortium — The Inevitability of the Combination of PCN × StunningBox The project requires 'AI platform technology' and 'professional translation and interpretation domain knowledge' simultaneously, which cannot be achieved by one or the other alone. The combination of PCN (Platform·SaaS·API) and StunningBox (AI Translation Model·1.1 million corpus·Regulatory Certification) is the only combination that enables achievement of goals within 24 months. The principal R&D institution has experience in numerous AI R&D projects, possessing its own AI ontology platform, and plays a role not only in mere platform implementation but also in optimizing and applying AI model outcomes to actual services. Particularly, by combining ontology-based knowledge structure with AI models to implement domain-specific inference functions and agent-based automation functions, the practical utility value of the AI technology is maximized.

Classification	PCN [Lead] — Technology·Platform	StunningBox [Joint-1] — AI Model·Data
Core Competencies	25-year IT/AI company (Revenue 27 billion KRW, 170 employees). SaaS platform·API integration·Cloud infrastructure construction·operation	Possession of a 15-year professional translation corporate dataset. Exclusively holds a business and regulatory parallel corpus of over 1.1 million sentence pairs. RAG patent registration. Completion of Multi-Agent Translation PoC. eCTD·CE·FCC regulatory document structuring technology Fine-tuning of
Handling in This Project	SaaS platform development and operation, API integration for malls·CRM·ERP, customer onboarding, cloud infrastructure (AWS), CI/CD pipeline, commercialization·marketing overall management	translation and interpretation AI models (NLLB/ EXAONE), E2E speech interpretation model (InfiniSST), RAG architecture design, regulatory certification formatting module, data cleansing and revision
Synergy upon Combination	Installed StunningBox's specialized AI models on PCN's proven platform infrastructure → Development period reduced by 32 months, costs reduced by 340 million KRW	The model trained on StunningBox's 1.1 million corpus is distributed to all industries via PCN SaaS platform → Data construction 8 months reduced

(2) Commercialization Organization Composition Chart

Organization	Participating Personnel	Assigned Technology Details
(주)피씨엔 [주관]	총괄책임자 외 10명 (총 11명)	SaaS 플랫폼 개발·통합, API 연동(쇼핑몰·CRM), 클라우드 인프라 구축, 수요기업 온보딩, 사업화 총괄
(주)스터닝박스 [공동-①]	공동책임자 외 6명 (총 7명)	AI 번역·통역 모델 파인튜닝, E2E 음성통역, RAG 아키텍처, 규제인증 포맷화, 학습 데이터 정제·감수
수요기업 3개사 [공동-②]	기업당 담당자 2명 (총 6명)	업종별 현장 데이터 제공, 솔루션 실증·검증, 도입 효과 측정 협조, 피드백 제공

2) Researcher Status (Lead Organization - PCN)

Name	Affiliation·Position	Education·Major	Role during Project Execution
Um Sang-hyun	PCN Director	Master/AI·SW	Research Lead. Overall platform development responsibility, commercialization strategy establishment, consortium coordination
Choi Beom-jin	PCN Director	Bachelor/SW Engineering	Establishment and advancement of domain knowledge-based ontology
Lee Kyung-soo	PCN Manager	Bachelor/ Backend	SaaS platform architecture design, cloud infrastructure construction and operation
Kim Jung-woo	PCN Assistant Manager	Bachelor/ AI·MLOps	AI model deployment and serving (vLLM), CI/CD automation, performance monitoring
Choi Seung-bum	PCN Manager	Master/AI·SW	Design and implement application AI logic (agent, automation functions, etc.)
Lee Hye-jung	PCN Chief	Master/AI·SW	Customer onboarding automation, dashboard development, user training and customer support
Lee Jae-man	PCN Manager	Bachelor/Data	Data collection and cleansing for customers, RAG vector DB construction, onboarding support
Hong Hyo-jin	PCN Chief	Bachelor/ Backend	SaaS platform architecture design, cloud infrastructure construction and operation
Kim Tae-heon	PCN Team Leader	Bachelor/QA	System testing, quality assurance, TTA/KAIC official verification response
Park Seo-ri	PCN Assistant	Bachelor/AI	Design and implement application AI logic (agent, automation functions, etc.)
Seo Min-gyu	PCN Assistant Manager	Bachelor/ Security	Cloud security, personal information anonymization, SLA monitoring
New Hire ①	PCN Researcher	Bachelor/Data	Data collection and cleansing for customers, RAG vector DB construction, onboarding support
New Hire ②	PCN Researcher	Bachelor/ Security	Cloud security, personal information anonymization, SLA monitoring
New Hire ③	PCN Researcher	Bachelor/QA	System testing, quality assurance, TTA/KAIC official verification response
New Hire ④	PCN Researcher	Bachelor/AI	Design and implement application AI logic (agent, automation functions, etc.)
New Hire ⑤	PCN Researcher	Bachelor/SW	Customer onboarding automation, dashboard development, user training and customer support
New Hire ⑥	PCN Researcher	Bachelor/ Marketing	Commercialization and marketing execution, KOTRA·Trade Association linkage, customer management

3) Researcher Status of Participating Institutions - StunningBox)

Name	Affiliation·Position	Education·Major	Role during Project Execution
Jeong Sumin	Stunning Box Co., Ltd. Manager	Bachelor's/Expert in Data 15 years	Co-Principal Investigator. Quality supervision and construction of 1.1 million corpus Overall supervisor, Co-inventor of RAG Patent
Eom Junhyeon	Stunning Box Co., Ltd. Manager	Bachelor's/ Developer 15 years	AI Model Fine-tuning Design, E2E Speech Interpretation Model Architecture, RAG System Design
Yun Inhee	Stunning Box Co., Ltd.	Master's/Planner 5 years	Quality management of regulatory documents (eCTD·CE·FCC) and planning the overall flow of solutions
Kang Hyojeong	Stunning Box Co., Ltd. Manager	Bachelor's/ Development	AI Model Quality Verification, Translation Result Evaluation (BLEU·MQM), Field Verification Support and Feedback Collection from Demand Companies
Ma Jaemin	Stunning Box Co., Ltd. Manager	Bachelor's/ Translation·Data	Dataset Construction·Labeling, Standardization of Technical Terminology Dictionaries, eCTD XML Structuring·PDF/A Conversion
Lee Junu	Stunning Box Co., Ltd. Manager	Bachelor's/Regulatory Data Supervision	LLM Fine-tuning·RLHF Training / Quality Management of Regulatory Documents (eCTD·CE·FCC)
Jo Min	Stunning Box Co., Ltd. Manager	Bachelor's/UX Design	UI/UX Design
New Hire ①	Stunning Box Co., Ltd. Researcher	Master's/ Translation Review and Expert Usability Inspection	Dataset Construction·Labeling, Standardization of Technical Terminology Dictionaries, eCTD XML Structuring·PDF/A Conversion
New Hire ②	Stunning Box Co., Ltd. Researcher	Master's/AI·NLP	LLM Fine-tuning·RLHF Training Practice, E2E Speech Interpretation Model Training, Model Performance Evaluation·Optimization
New Hire ③	Stunning Box Co., Ltd. Researcher	Bachelor's/Data Engineering	Corpus Refinement Pipeline Construction, RAG Vector DB Embedding·Indexing, Data Quality Verification (IAA 0.85+)
New Hire ④	Stunning Box Co., Ltd. Researcher	Master's/ Translation Review and Expert Usability Inspection	Development of Regulatory Certification Formatting Module (eCTD·CE·FCC·HACCP), Document Auto-generation Agent Backend Development

4) Patents, Papers, Awards Related to This Project

(1) Three Demand Companies — Demonstration Verification Participation Capability

Demand Companies	Industry	Export Performance·Capability	Contents of Demonstration Participation	Contribution to This Project (Data·Verification)
Sunlight Ilucom Co., Ltd.	Automotive Lighting Parts	Audi/VW Tier1 Registration (2017). $30 million Export Certification (2024). Ulsan New Factory (2025)	German·English·Japanese Translation Verification. Buyer Consultation Interpretation Verification	Provision of Technical Specification·Quality Criteria·OEM Contract·Exhibition Script. Quantitative Measurement of Time·Cost Reduction
Genuine Grip Co., Ltd.	Special Shoes Manufacturing	Expert in Export and Customs Clearance. Numerous Buyer Consultations·Exhibition Participations	Verification of English·Chinese·Vietnamese Translation. Multilingual Catalog	Provision of Buyer Negotiation Dialogues·Product Catalog Specifications·Raw Material Manuals. Exhibition Interpretation On-site Verification
Gray Whale	K-Beauty Cosmetics	CE Marking·China NMPA Export. K-Beauty Global Marketing	Verification of English·Chinese·Japanese·French Translation. Regulatory Certification Document Verification	Provision of CE·FDA Component Documents·Marketing Content·Regulatory Submission Documents. Measurement of Automated Certification Document Effectiveness

Three PCN Patents (Ontology Search Patent, OASIS GS Certification, Oz-i Platform) + Stunning Box Two Patents (RAG Patent, 1.1 million Corpus) — Refer to Chapter 3, Previous Research and Development for Details ※ This consortium is the only combination that embodies the 'Technology (PCN Platform) × Data (Stunning Box 1.1 million Corpus) × Market (4 Demand Industries)', which is the key foundation to achieving TRL 4→7 and commercialization within 24 months.

5. Technology Development Schedule

1) Overall Development Schedule (2026.7 ~ 2028.6, Total 24 months)

■ PCN Responsibility ■ Stunning Box Responsibility ■ Joint Execution ■ Demand Company Participation

No.	Development Content	담당	7~8	9~10	11~12	1~2	3~4	5~6	7~8	9~10	11~12	1~2	3~4	5~6
			First Year (2026.7~2027.6)			Second Year (2027.7 ~2028.6)								
1	Requirements Definition ·Architecture Design	Joint	●	●										
2	Business·Regulatory Corpus Collection ·Refinement ·Term DB	Stunning	●	●	●									
3	Field Data Collection from Three Demand Companies ·Anonymization	Joint	●	●	●	●								
4	Business Translation Model Fine-tuning (NLLB/ EXAONE)	Stunning		●	●	●	●							
5	RAG Corporate Knowledge DB · Hybrid Search Engine	Stunning			●	●	●							
6	E2E Streaming Speech Interpretation Model (InfiniSST)	Stunning			●	●	●	●	●					
7	SaaS Platform ·Cloud Infrastructure Construction	PCN	●	●	●	●	●	●						
8	API Integration (Cafe24·Amazon ·CRM·ERP)	PCN				●	●	●	●					
9	Five Types of Export Document Auto-generation Agents	Joint				●	●	●	●	●				
10	Regulatory Certification Document Formatting Module (eCTD·CE·FCC)	Stunning					●	●	●	●	●			
11	Web·Mobile UI/UX ·	PCN						●	●	●	●			

	Zoom/Teams Plugin														
12	RLHF Reinforcement Learning·Model Enhancement	Stunning						●	●	●					
13	Alpha Test (Client Company 3 Firms Internal)	Joint					●	●							
14	Beta Test (Client Company 3 Firms Practical Application)	Joint							●	●	●				
15	CI/CD Automation · Monitoring Da shboard	PCN								●	●	●			
16	Empirical Validation (3 Firms Full · A/B Comparison · Performance Measurement)	Joint										●	●		
17	TTA/KAIC Official Verification · Co mmercialization Preparation	Joint											●	●	

2) Detailed Role Distribution by Institution

Responsibility	Development Items	Detailed Execution Content	Deliverables
PCN	SaaS Platform · Cloud Infrastructure	Oz-i Based Multi-Tenant SaaS Architecture Construction. AWS Cloud Infrastructure (EKS, RDS, S3). 99.9% Availability SLA Operating System	SaaS Platform v1.0, Cloud In frastructure Operation Manual
PCN	API Integration Module	OASIS Architecture Based RESTful API Gateway. Cafe24·Amazon·Naver Shopping Mall Integration. CRM·ERP Integration Module	API Gateway, Shopping Mall Integration SDK, API Documentation
PCN	Web·Mobile UI/UX · Plugin	React Based Dashboard. Client Company Onboarding UI. Zoom·MS Teams Native Pluggin. Mobile App (iOS·Android)	Web App v1.0, Mobile App v1.0, Zoom Pl ugin
PCN	CI/CD · MLOps · Monitoring	Kubernetes Auto Scaling. CI/CD Pipeline (GitHub Actions). Prometheus+Grafana Monitoring. Auto Relearning Pipeline	CI/CD Pipeline, Monitoring Dashboard, SLA Report
StunningBox	Corpus Collection · Refinement · Terminology DB	1.1 million+ sentence pairs cleansing·de-identification·labeling. Industry-specific dictionary construction. Data quality validation (IAA 0.85+)	Refined Corpus 1.1 Million+, 3 Types of Terminology Dictionary, Quality Report
StunningBox	Translation Model Fine Tuning · RLHF	NLLB/EXAONE Domain Fine-Tuning (LoRA). Three industry-specific adapters. DPO-based RLHF. Translation Expert Feedback Loop	Translation Model v1.0~v3.0, 3 Industry-Specific Adapters
StunningBox	E2E Speech Interpretation Model	InfiniSST: Whisper+Wav2Vec2+Qwen2.5-7B. AlignAtt Policy. Sliding KV-Cache. 8 + Languages Bidirectional	E2E Interpretation Model v1.0, Streaming API
StunningBox	RAG Architecture · Regulatory Certification Formatting	RAG Patent-based Hybrid Search Engine. eCTD·CE·FCC·HACCP Automatic Formatting. XML/PDF/A Auto Conversion	RAG Engine v1.0, Regulatory Certification Formatting Module v1.0
Joint	Client Company Empirical · Validation	Alpha→Beta→Empirical 3-phase Verification. A/B Comparison Measurement. Work Hours·Cost Reduction Quantitative Data Collection	Alpha·Beta Test Report, Empirical Performance Report

Joint	TTA/KAIC Official Verification · Co mmercialization	Official Agency Test Request→Execution→Certificate Issuance. AI Performance+SW Quality+Service Stability Comprehensive Verification. Commercialization SaaS Launch Preparation	Official Certification, Commercialization SaaS v1.0, Business Plan

3) Key Milestones by Year

Timeline	Milestone	Achievement Criteria (Go/No-Go)
1차년도 ('26.12)	코퍼스 정제 완료, 통번역 모델 v1.0, E2E 프로토타입, SaaS 인프라 구축	BLEU 55+, WER 20%↓, 통역 지연 3초↓. 미달 시 데이터 보강 후 재평가
1차년도 ('27.6)	알파 테스트 완료, API 연동 완료, 문서 에이전트 3종, 모델 v2.0	BLEU 60+, WER 17%↓, 알파 기능 정상률 90%+. Go: 베타 진입
2차년도 ('27.12)	베타 테스트 완료, E2E 2초 달성, 규제인증 모듈 v1.0, 에이전트 5종 완성	BLEU 63+, StreamLAAL 2.5초↓, 만족도 3.5/5.0+. Go: 실증 진입
Second Year ('28.06)	Empirical Validation Completed, TTA/KAIC Official Certification, SaaS v1.0 Commercialization Launch	BLEU 65+, WER 15%↓, StreamLAAL 2 Seconds↓, F1 0.8+, TEDS 70%+. Work Day Reduced by 2 Hours+

6. Research Expense Use Plan, etc.

1. Research and Development Cost Use Plan

1) Research and Development Cost Support and Burden Plan

(Unit: Thousand Won)

Classification			Government-Supported Research and Development Cost	Institution-Burden Research and Development Cost			Total		
Year	Institution Role 1」	Research and Development Institution Name	Cash	Cash	In-kind	Subtotal	Cash	In-kind	Total
1	Principal	PCN Co., Ltd.	312,500	10,425	93,825	104,250	322,925		416,750
1	Participant	StunningBox Co., Ltd.	312,500	10,425	93,825	104,250	322,925		416,750
2	Principal	PCN Co., Ltd.	625,000	20,850	187,650	208,500	645,850		833,500
2	Participant	StunningBox Co., Ltd.	625,000	20,850	187,650	208,500	645,850		833,500
3	Principal	PCN Co., Ltd.	312,500	10,425	93,825	104,250	322,925		416,750
3	Participant	StunningBox Co., Ltd.	312,500	10,425	93,825	104,250	322,925		416,750
Grand Total			2,500,000	83,400	750,600	834,000	2,583,400		3,334,000

2) Research and Development Cost Use Plan

(1) Research and Development Cost Use Plan by Institution

(Unit: Thousand Won)

Research and Development Institution Name		Research and Development Cost													Research Development Cost Other Support Fund5」	Research Inducement Calculation Standard Amount6」
		Direct Cost											Indirect Cost	Total		
		Labor Cost	Student Labor Cost		Research Facility and Equipment Cost		Research Material Cost	위탁연구개발비	국제공동연구개발비	연구개발부담비	Research Activity Expenses	연구수당	Subtotal			
			General 1」	Special Case 2」	General 3」	Special Case 4」										
PCN Co., Ltd.	Cash	916,330									355,370		1,271,700	20,000	1,291,700	
	In-kind	375,300											375,300		375,300	
	Subtotal	1,291,630									355,370		1,647,000	20,000	1,667,000	
Corporation Stunning Box	Cash	784,000					55,000				406,700		1,245,700	46,000	1,291,700	
	In-kind	375,300											375,300		375,300	
	Subtotal	1,159,300					55,000				406,700		1,621,000	46,000	1,667,000	
Grand Total	Cash	1,700,330					55,000				762,070		2,517,400	66,000	2,583,400	
	In-kind	750,600											750,600		750,600	
	Total	2,450,930					55,000				762,070		3,268,000	66,000	3,334,000	

(2) Annual Usage Plan

(Unit: Thousand Won)

Year		Research and Development Cost													Research Development Cost Other Support Fund5」	Research Inducement Calculation Standard Amount6」
		Direct Cost											Indirect Cost	Total		
		Labor Cost	Student Labor Costs		Research Facilities and Equipment Expenses		Research Material Cost	위탁연구개발비	국제공동연구개발비	연구개발부담비	Research Activity Expenses	Research Allowance	Subtotal			
			General	Special Case	General	Special Case										
1	Cash	405,330					20,000				208,520		633,850	12,000	645,850	
	In-kind	187,650											187,650		187,650	
	Subtotal	592,980					20,000				208,520		821,500	12,000	833,500	
2	Cash	863,000					25,000				370,700		1,258,700	33,000	1,291,700	
	In-kind	375,300											375,300		375,300	
	Subtotal	1,238,300					25,000				370,700		1,634,000	33,000	1,667,000	
3	Cash	432,000					10,000				182,850		624,850	21,000	645,850	
	In-kind	187,650											187,650		187,650	
	Subtotal	619,650					10,000				182,850		812,500	21,000	833,500	
Grand Total	Cash	1,700,330					55,000				762,070		2,517,400	66,000	2,583,400	
	In-kind	750,600											750,600		750,600	
	Total	2,450,930					55,000				762,070		3,268,000	66,000	3,334,000	

(3) Annual Usage Plan by Research and Development Institution

A. Name of Lead Research and Development Institution: PCN Co., Ltd.

(Unit: Thousand Won)

Year		Research and Development Cost														Research Development Cost Other Support Fund5」	Research Inducement Calculation Standard Amount6」
		Direct Cost													Indirect Cost	Total	
		Labor Cost	Student Labor Costs		Research Facilities and Equipment Expenses		Research Material Cost	위탁연구개발비	국제공동연구개발비	연구개발부담비	Research Activity Expenses	Research Allowance	Subtotal			
			General	Special Case	General	Special Case										
1	Cash	207,830									109,095		316,925	6,000	322,925		
	In-kind	93,825											93,825		93,825		
	Subtotal	301,655									109,095		410,750	6,000	416,750		
2	Cash	474,000									163,850		637,850	8,000	645,850		
	In-kind	187,650											187,650		187,650		
	Subtotal	661,650									163,850		825,500	8,000	833,500		
3	Cash	234,500									82,425		316,925	6,000	322,925		
	In-kind	93,825											93,825		93,825		
	Subtotal	328,325									82,425		410,750	6,000	416,750		
Grand Total	Cash	916,330									355,370		1,271,700	20,000	1,291,700		
	현물	375,300											375,300		375,300		
	Total	1,291,630									355,370		1,647,000	20,000	1,667,000		

o Status Table of Participating Researchers' Labor Costs (First Year) ※ Compiled yearly for the entire project period

Personnel Classification	Name	Position (Indicate Student if applicable)	Internal/ External and Support Classification*	New/ Existing	Total Salary (A)	Participation Period (B, months)	Participation Rate (%) (C)	Total ((A/12)×B×(C/100))		
								Cash	In-kind	Total
Participating Researcher (Existing)	Choi Beom-jin	Director	Internal	Existing	173,588	6	10		8,600	8,600
	Kim Jung-woo	Manager	Internal	Existing	61,503	6	20		6,105	6,105
	Lee Kyung-soo	Department Head	Internal	Existing	87,400	6	15		6,550	6,550
	Choi Seung-bum	Department Head	Internal	Existing	146,500	6	11		8,050	8,050
	Lee Hye-jung	Director	Internal	Existing	146,500	6	11		8,050	8,050
	Lee Jae-man	Department Head	Internal	Existing	80,000	6	30		12,000	12,000
	Hong Hyo-jin	Director	Internal	Existing	150,000	6	19		14,200	14,200
	Kim Tae-heon	Group Leader	Internal	Existing	99,100	6	28		13,870	13,870
	Park Seo-ri	Deputy Manager	Internal	Existing	65,600	6	50		16,400	16,400
	Subtotal (D)								93,825	93,825
Participating Researcher (New)	Um Sang-hyun	Director	Internal	New	70,000	6	100	35,000		35,000
	Seo Min-gyu	Manager	Internal	New	49,000	6	100	24,500		24,500
	Scheduled Hire 1	–	Internal	New (Youth 1)	50,000	5	100	20,833		20,833
	Scheduled Hire 2	–	Internal	New (Youth 2)	50,000	5	100	20,833		20,833
	Scheduled Hire 3	–	Internal	New	50,000	5	100	20,833		20,833
	Scheduled Hire 4	–	Internal	New	50,000	5	100	20,833		20,833
	Scheduled Hire 5	–	Internal	New	50,000	5	100	20,833		20,833
	Scheduled Hire 6	–	Internal	New	50,000	5	100	20,833		20,833
	Scheduled Hire 7	–	Internal	New	35,000	4	100	11,666		11,666
	Scheduled Hire 8	–	Internal	New	35,000	4	100	11,666		11,666
	Subtotal (E)							207,830		207,830
Total (G=D+E+F)								207,830	93,825	301,655

o Status Table of Participating Researchers' Labor Costs (Second Year) ※ Compiled yearly for the entire project period

Personnel Classification	Name	Position (Indicate Student if applicable)	Internal/ External and Support Classification*	New/ Existing	Total Salary (A)	Participation Period (B, months)	Participation Rate (%) (C)	Total ((A/12)×B×(C/100))		
								Cash	In-kind	Total
Participating Researcher (Existing)	Choi Beom-jin	Director	Internal	Existing	173,588	12	10		17,000	17,000
	Kim Jung-woo	Manager	Internal	Existing	61,503	12	20		12,300	12,300
	Lee Kyung-soo	Department Head	Internal	Existing	87,400	12	15		13,110	13,110
	Choi Seung-bum	Department Head	Internal	Existing	146,500	12	11		16,100	16,100
	Lee Hye-jung	Director	Internal	Existing	146,500	12	11		16,100	16,100
	Lee Jae-man	Department Head	Internal	Existing	80,000	12	30		24,000	24,000
	Hong Hyo-jin	Director	Internal	Existing	150,000	12	19		28,500	28,500
	Kim Tae-heon	Group Leader	Internal	Existing	99,100	12	28		27,740	27,740
	Park Seo-ri	Deputy Manager	Internal	Existing	65,600	12	50		32,800	32,800
	Subtotal (D)								187,650	187,650
Participating Researcher (New)	Um Sang-hyun	Director	Internal	New	70,000	12	100	70,000		70,000
	Seo Min-gyu	Manager	Internal	New	49,000	12	100	49,000		49,000
	Scheduled Hire 1	–	Internal	New (Youth 1)	50,000	12	100	50,000		50,000
	Scheduled Hire 2	–	Internal	New (Youth 2)	50,000	12	100	50,000		50,000
	Scheduled Hire 3	–	Internal	New	50,000	12	100	50,000		50,000
	Scheduled Hire 4	–	Internal	New	50,000	12	90	45,000		45,000
	Scheduled Hire 5	–	Internal	New	50,000	12	90	45,000		45,000
	Scheduled Hire 6	–	Internal	New	50,000	12	90	45,000		45,000
	Scheduled Hire 7	–	Internal	New (Youth 3)	35,000	12	100	35,000		35,000
	Scheduled Hire 8	–	Internal	New (Youth 4)	35,000	12	100	35,000		35,000
	Subtotal (E)							474,000		474,000
Total (G=D+E+F)								474,000	187,650	661,650

o Status Table of Participating Researchers' Labor Costs (Third Year) ※ Compiled yearly for the entire project period

Personnel Classification	Name	Position (Indicate Student if applicable)	Internal/ External and Support Classification*	New/ Existing	Total Salary (A)	Participation Period (B, months)	Participation Rate (%) (C)	Total ((A/12)×B×(C/100))		
								Cash	In-kind	Total
Participating Researcher (Existing)	Choi Beom-jin	Director	Internal	Existing	173,588	6	10		8,600	8,600
	Kim Jung-woo	Manager	Internal	Existing	61,503	6	20		6,105	6,105
	Lee Kyung-soo	Department Head	Internal	Existing	87,400	6	15		6,550	6,550
	Choi Seung-bum	Department Head	Internal	Existing	146,500	6	11		8,050	8,050
	Lee Hye-jung	Director	Internal	Existing	146,500	6	11		8,050	8,050
	Lee Jae-man	Department Head	Internal	Existing	80,000	6	30		12,000	12,000
	Hong Hyo-jin	Director	Internal	Existing	150,000	6	19		14,200	14,200
	Kim Tae-heon	Group Leader	Internal	Existing	99,100	6	28		13,870	13,870
	Park Seo-ri	Deputy Manager	Internal	Existing	65,600	6	50		16,400	16,400
	Subtotal (D)								93,825	93,825
Participating Researcher (New)	Um Sang-hyun	Director	Internal	New	70,000	6	100	35,000		35,000
	Seo Min-gyu	Manager	Internal	New	49,000	6	100	24,500		24,500
	Scheduled Hire 1	–	Internal	New (Youth 1)	50,000	6	100	25,000		25,000
	Scheduled Hire 2	–	Internal	New (Youth 2)	50,000	6	100	25,000		25,000
	Scheduled Hire 3	–	Internal	New	50,000	6	90	22,500		22,500
	Scheduled Hire 4	–	Internal	New	50,000	6	90	22,500		22,500
	Scheduled Hire 5	–	Internal	New	50,000	6	90	22,500		22,500
	Scheduled Hire 6	–	Internal	New	50,000	6	90	22,500		22,500
	Scheduled Hire 7	–	Internal	New (Youth 3)	35,000	6	100	17,500		17,500
	Scheduled Hire 8	–	Internal	New (Youth 4)	35,000	6	100	17,500		17,500
	Subtotal (E)							234,500		234,500
Total (G=D+E+F)								234,500	93,825	328,325

B. Name of Joint Research and Development Institution: Stunning Box

(Unit: Thousand Won)

Year		Research and Development Cost													Indirect Cost	Total	Support Fund Outside Research Development Expenses 5」	Research Inducement Calculation Standard Amount6」
		Direct Cost																
		Labor Cost	Student Labor Costs		Research Facilities and Equipment Expenses		Research Material Cost	위탁연구개발비	International Joint Research and Development Expenses	연구개발부담비	Research Activity Expenses	Research Allowance	Subtotal					
			General	Special Case	General	Special Case												
1	Cash	197,500					20,000				99,425		316,925	6,000	322,925			
	In-kind	93,825											93,825		93,825			
	Subtotal	291,325					20,000				99,425		410,750	6,000	416,750			
2	Cash	389,000					25,000				206,850		620,850	25,000	645,850			
	In-kind	187,650											187,650		187,650			
	Subtotal	576,650					25,000				206,850		808,500	25,000	833,500			
3	Cash	197,500					10,000				100,425		307,925	15,000	322,925			
	In-kind	93,825											93,825		93,825			
	Subtotal	291,325					10,000				100,425		401,750	15,000	416,750			
Grand Total	Cash	784,000					55,000				406,700		1,245,700	46,000	1,291,700			
	In-kind	375,300											375,300		375,300			
	Total	1,159,300					55,000				406,700		1,621,000	46,000	1,667,000			

o Status Table of Participating Researchers' Labor Costs (First Year) ※ Compiled yearly for the entire project period

Personnel Classification	Name	Position (Indicate Student if applicable)	Internal/ External and Support Classification*	New/ Existing	Total Salary (A)	Participation Period (B, months)	Participation Rate (%) (C)	Total ((A/12)×B×(C/100))		
								Cash	In-kind	Total
Participating Researcher (Existing)	Jeong Sumin	CEO	Internal	Existing	110,000	6	98.9		54,395	54,395
	Eom Junhyeon	Director	Internal	Existing	40,000	6	98.6		19,720	19,720
	Ma Jaemin	Manager	Internal	Existing	40,000	6	98.55		19,710	19,710
	Subtotal (D)								93,825	93,825
Participating Researcher (New)	Lee Junu	Department Head	Internal	New	60,000	6	100	30,000		30,000
	Cho Min	Department Head	Internal	New	40,000	6	100	20,000		20,000
	Kang Hyojeong	Manager	Internal	New	45,000	6	100	22,500		22,500
	Yun Inhee	Manager	Internal	New	45,000	6	100	22,500		22,500
	Scheduled Hire 1	Director	Internal	New	84,000	5	100	35,000		35,000
	Scheduled Hire 2	Department Head	Internal	New	81,000	5	100	33,750		33,750
	Scheduled Hire 3	Manager	Internal	New	55,000	5	100	22,916		22,916
	Scheduled Hire 4	Deputy Manager	Internal	New (Youth 5)	26,000	5	100	10,833		10,833
	Subtotal (E)							197,500		291,325
Total (G=D+E+F)								197,500	93,825	291,325

o Status Table of Participating Researchers' Labor Costs (Second Year) ※ Compiled yearly for the entire project period

Personnel Classification	Name	Position (Indicate Student if applicable)	Internal/ External and Support Classification*	New/ Existing	Total Salary (A)	Participation Period (B, months)	Participation Rate (%) (C)	Total ((A/12)×B×(C/100))		
								Cash	In-kind	Total
Participating Researcher (Existing)	Jeong Sumin	CEO	Internal	Existing	110,000	12	98.9		108,790	108,790
	Eom Junhyeon	Director	Internal	Existing	40,000	12	98.6		39,440	39,440
	Ma Jaemin	Manager	Internal	Existing	40,000	12	98.55		39,420	39,420
	Subtotal (D)								187,650	187,650
Participating Researcher (New)	Lee Junu	Department Head	Internal	New	60,000	12	53	31,800		31,800
	Cho Min	Department Head	Internal	New	40,000	12	53	21,200		21,200
	Kang Hyojeong	Manager	Internal	New	45,000	12	100	45,000		45,000
	Yun Inhee	Manager	Internal	New	45,000	12	100	45,000		45,000
	Scheduled Hire 1	Director	Internal	New	84,000	12	100	84,000		84,000
	Scheduled Hire 2	Department Head	Internal	New	81,000	12	100	81,000		81,000
	Scheduled Hire 3	Manager	Internal	New	55,000	12	100	55,000		55,000
	Scheduled Hire 4	Deputy Manager	Internal	New (Youth 5)	26,000	12	100	26,000		26,000
	Subtotal (E)							389,000		576,650
Total (G=D+E+F)								389,000	187,650	576,650

o Status Table of Participating Researchers' Labor Costs (Third Year) ※ Compiled yearly for the entire project period

Personnel Classification	Name	Position (Indicate Student if applicable)	Internal/ External and Support Classification*	New/ Existing	Total Salary (A)	Participation Period (B, months)	Participation Rate (%) (C)	Total ((A/12)×B×(C/100))		
								Cash	In-kind	Total
Participating Researcher (Existing)	Jeong Sumin	CEO	Internal	Existing	110,000	6	98.9		54,395	54,395
	Eom Junhyeon	Director	Internal	Existing	40,000	6	98.6		19,720	19,720
	Ma Jaemin	Manager	Internal	Existing	40,000	6	98.55		19,710	19,710
	Subtotal (D)								93,825	93,825
Participating Researcher (New)	Lee Junu	Department Head	Internal	New	60,000	6	59	17,700		17,700
	Cho Min	Department Head	Internal	New	40,000	6	59	11,800		11,800
	Kang Hyojeong	Manager	Internal	New	45,000	6	100	22,500		22,500
	Yun Inhee	Manager	Internal	New	45,000	6	100	22,500		22,500
	Scheduled Hire 1	Director	Internal	New	84,000	6	100	42,000		42,000
	Scheduled Hire 2	Department Head	Internal	New	81,000	6	100	40,500		40,500
	Scheduled Hire 3	Manager	Internal	New	55,000	6	100	27,500		27,500
	Scheduled Hire 4	Deputy Manager	Internal	New (Youth 5)	26,000	6	100	13,000		13,000
	Subtotal (E)							197,500		291,325
Total (G=D+E+F)								197,500	93,825	291,325

2. Detailed Usage Plan for Research and Development Expenses

(1) Research Facilities and Equipment Construction Plan - Not Applicable

(2) Purchase Costs for Research Materials (Reagents, Materials, etc.)

Research and Development Institution	Purchase Year	Research Materials	Purpose of Use	Purchase Quantity (g)	Purchase Cost (thousand won)	Purchase (Receipt) Date	Usage Destination
Stunning Box Co., Ltd.	Year 2026	Domain-Specific AI Learning Global Trade and Export Company Dataset	Establishment and Fine-tuning of Basic Trade and Export Specialized Data for AI Translation and Analysis Model Learning in the First Year, Global Customs, Export Regulation, and Buyer Specialized Dataset	1 set	20,000	August 2026	AI Model Research Laboratory / Training Server
	2027				25,000	January 2027	
	2028				10,000	January 2028	

(2-1) Research Material Production Costs (Test Products, Facilities, etc.) Usage Plan - Not Applicable

(3) Research Activity Costs (Outsourcing Fees)

Research and Development Institution	Year	Service Name	Service Description	Service Company	Required Costs (thousand KRW)	Service Period	Deliverables
㈜0000	2026						
	2026						

(3-1) Research Activity Costs (Intellectual Property Creation Activities, External Expert Technology Usage Fees, Software Usage Fees, Laboratory Operation Expenses, Comprehensive Management Fees, Travel Expenses, and Other Costs)

Research and Development Institution	Year	Name (Category)	Method	Usage (Company)	Quantity (Cases)	Required Costs (thousand KRW)	Usage (Execution) Period	Installation Location (Deliverables)
PCN Co., Ltd.	2026	Entrusted Settlement Fees (Other Costs)	Purchase	Settlement Fees for Year 2026	1	3,000	'26.7 ~12	PCN
		Cloud Usage Fees (Cloud Computing Service Usage Fees)	Rental	Cloud Usage Fees	6	50,000	'26.7 ~12	PCN
		Office Supplies Expenses (Laboratory Operation Costs)	Purchase	Office Supplies Purchase	5	2,000	'26.7 ~12	PCN
		Test Certification Fees (External Expert Technology Usage Fees)	Purchase	Test Analysis and Certification	2	5,000	'26.7 ~12	PCN
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	Technical Consultation	2	20,000	'26.7 ~12	PCN
		Conference and Seminar Participation Fees (Research Personnel Support Fees)	Purchase	Academic Conference Participation Fees	4	15,000	'26.7 ~12	PCN
		Overseas Travel Expenses (Travel Expenses)	Purchase	Overseas Travel Expenses	1	10,000	'26.7 ~12	PCN
		Domestic Travel Expenses (Travel Expenses)	Purchase	Domestic Travel Expenses	10	1,670	'26.7 ~12	PCN
	2027	Entrusted Settlement Fees (Other Costs)	Purchase	Settlement Fees for Year 2027	1	3,000	'27.1 ~12	PCN
		Cloud Usage Fees (Cloud Computing Service Usage Fees)	Rental	Cloud Usage Fees	12	85,000	'27.1 ~12	PCN
		Office Supplies Expenses (Laboratory Operation Costs)	Purchase	Office Supplies Purchase	5	2,000	'27.1 ~12	PCN
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	Technical Consultation	2	20,000	'26.7 ~12	PCN
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	Model Verification	1	10,000	'27.1 ~12	PCN
		Conference and Seminar Participation Fees (Research Personnel Support Fees)	Purchase	Conference Presentation	1	10,000	'27.1 ~12	PCN
		Overseas Travel Expenses (Travel Expenses)	Purchase	Overseas Travel Expenses	1	10,000	'26.7 ~12	PCN
		Domestic Travel Expenses (Travel Expenses)	Purchase	Domestic Travel Expenses	10	6,850	'27.1 ~12	PCN
	2028	Entrusted Settlement Fees (Other Costs)	Purchase	Settlement Fees for Year 2028	1	3,000	'28.1 ~6	PCN
		Cloud Usage Fees (Cloud Computing Service Usage Fees)	Rental	Cloud Usage Fees	6	30,000	'28.1 ~6	PCN
		Office Supplies Expenses (Laboratory Operation Costs)	Purchase	Office Supplies Purchase	5	2,000	'28.1 ~6	PCN
		Test Certification Fees (External Expert Technology Usage Fees)	Purchase	Test Analysis and Certification	2	5,000	'28.1 ~6	PCN
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	UX Testing	2	30,000	'28.1 ~6	PCN
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	Promotional Video Production Service	1	5,000	'28.1 ~6	PCN
		Conference and Seminar Participation Fees (Research Personnel Support Fees)	Purchase	Workshop Hosting Costs		5,000	'28.1 ~6	PCN

		Domestic Travel Expenses (Travel Expenses)	Purchase	Domestic Travel Expenses	10	1,425	'28.1 ~6	PCN
Stunning Box Co., Ltd.	2026	Cloud Usage Fees (Cloud Computing Service Usage Fees)	Rental	Translation/Interpretation Model Training/Inference Utilization GPU Cloud	6	60,000	'26.7 ~12	StunningBox
		Office Supplies Expenses (Laboratory Operation Costs)	Purchase	Office Supplies Purchase	-	2,000	'26.7 ~12	StunningBox
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	Domain-Specific Review Expert	-	23,000	'26.7 ~12	StunningBox
		Conference and Seminar Participation Fees (Research Personnel Support Fees)	Purchase	AI Technology Seminar	5	3,925	'26.7 ~12	StunningBox
		Overseas Travel Expenses (Travel Expenses)	Purchase	Overseas Travel Expenses	10	3,000	'26.7 ~12	StunningBox
		Domestic Travel Expenses (Travel Expenses)	Purchase	Domestic Travel Expenses	10	2,000	'26.7 ~12	StunningBox
		Meeting Costs (Meeting Costs)	Purchase	Meeting	10	1,500	'26.7 ~12	StunningBox
		Report Costs (Other Costs)	Purchase	Report Binding, Printing, etc.	10	3,000	'26.7 ~12	StunningBox
	2027	Cloud Usage Fees (Cloud Computing Service Usage Fees)	Rental	Translation/Interpretation Model Training/Inference Utilization GPU Cloud	12	130,000	'27.1 ~12	StunningBox
		Office Supplies Expenses (Laboratory Operation Costs)	Purchase	Office Supplies Purchase	5	2,000	'27.1 ~12	StunningBox
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	Domain-Specific Review Expert	-	33,000	'27.1 ~12	StunningBox
		Conference and Seminar Participation Fees (Research Personnel Support Fees)	Purchase	Researcher Training and Training Expenses	5	3,925	'27.1 ~12	StunningBox
		Domestic Travel Expenses (Travel Expenses)	Purchase	Domestic Travel Expenses	10	1,000	'27.1 ~12	StunningBox
		Meeting Costs (Meeting Costs)	Purchase	Meeting	10	1,500	'27.1 ~12	StunningBox
		Report Binding, Printing, etc. (Other Costs)	Purchase	Report Binding, Printing, etc.	10	3,000	'26.7 ~12	StunningBox
	2028	Cloud Usage Fees (Cloud Computing Service Usage Fees)	Rental	Translation/Interpretation Model Training/Inference Utilization GPU Cloud	6	62,000	'28.1 ~6	StunningBox
		Office Supplies Expenses (Laboratory Operation Costs)	Purchase	Office Supplies Purchase	5	1,000	'28.1 ~6	StunningBox
		Test Certification Fees (External Expert Technology Usage Fees)	Purchase	Test Analysis and Certification	2	15,000	'28.1 ~6	StunningBox
		Expert Usage Fees (External Expert Technology Usage Fees)	Purchase	Domain-Specific Review Expert	-	22,000	'28.1 ~6	StunningBox
		Conference and Seminar Participation Fees (Research Personnel Support Fees)	Purchase	Researcher Training and Training Expenses	3	5,425	'28.1 ~6	StunningBox
		Domestic Travel Expenses (Travel Expenses)	Purchase	Domestic Travel Expenses	5	1,000	'28.1 ~6	StunningBox
		Meeting Costs (Meeting Costs)	Purchase	Meeting	5	1,000	'28.1 ~6	StunningBox
		Report Binding, Printing, etc. (Other Costs)	Purchase	Report Binding, Printing, etc.	10	3,000	'28.1 ~6	StunningBox

(4) Indirect Costs

Research and Development Institution	Year	Details	Detailed Information	Required Costs (thousand KRW)
PCN Co., Ltd.	2026	Achievement Utilization Support Costs	Intellectual Property Rights Application and Registration	6,000
	2027	Achievement Utilization Support Costs	Intellectual Property Rights Application and Registration	8,000
	2028	Achievement Utilization Support Costs	Intellectual Property Rights Application and Registration	6,000
Stunning Box Co., Ltd.	2026	Achievement Utilization Support Costs	Intellectual Property Rights Application and Registration	6,000
	2027	Achievement Utilization Support Costs	Intellectual Property Rights Application and Registration	25,000
	2028	Achievement Utilization Support Costs	Intellectual Property Rights Application and Registration	15,000

3. Research Fund Procurement and Maintenance Plan

7. Utilization Plan and Expected Effects of R&D Achievements (Commercialization Goals and Strategies, etc.)

1) Commercialization/Product (Service) Goals and Plans

The commercialization goal of this project is to create a market centered on 'AI Translation SaaS', where SMEs can automate global sales, translation, and documentation without separate IT investment by linking their data. By occupying the empty position between existing general-purpose translation AI (DeepL, Google) and high-cost professional translation, it blocks entry into the competitive exclusive domain based on the Stanning Box 1.1 million+ corpus (including 500,000 regulatory certification).

(1) Stage-wise Commercialization Roadmap – Four-Stage Market Entry Strategy

Stage	Timing	Commercialization Goal	Key Execution Strategy	Sales Target	KPI
Stage 1 Base Construction	In R&D (~'28.6)	Pilot Completion of 3 Customer Companies. PMF Verification. Acquisition of TTA/KAIC Official Certification	Free Demonstration→ Securing Reference. Weaponization of Sales with Quantitative Introduction Effect Data	–	Adopted by 3 Companies, Daily Work Reduced by 2+ Hours, Certification Acquired
Phase 2 Commercialization	'28.7~ '29.6	Official Launch of SaaS. AI Voucher, KOTRA Channel Deployment. 50+ B2B Paid Clients	Simultaneous Deployment of Triple Channels (Government Support + Association + App Store). 30%+ Free Tier→Paid Conversion Rate	6 billion	50+ Paid Clients, MRR 500 million, NPS 50+
Stage 3 Expansion	'29.7~ '31.6	Opening of Industry-specific Marketplace. Cafe24 · Naver App Store. 15 Languages. 500+ Companies	Bulk Contracts with Industry Groups. Expansion of API B2B2B. Upselling to Existing Customers with Regulatory Certification Package	13 billion	10%+ B2B Market Share, LTV/CAC>3
Stage 4 Global	'31.7~ '33.6	Entry into Japan, Southeast Asia, and Middle East. Local CRO partners. eCTD Global. 1,000+ companies	API export model. Overseas localization (Japanese, Arabic, Vietnamese enhancement). On-premises enterprise	22 billion	Overseas sales 45%+, 1,000+ companies

(2) Product (Service) Line-up and Pricing Strategy

Product Name	Form	Price Range	Target Customers	Differentiation Points / Strategic Role
StunningBiz AI Starter	SaaS subscription (self-service)	100,000 KRW per month (1.2 million KRW/year)	Export beginner companies, Individual traders, E-commerce sellers	Natural transition from Free Tier (100 uses/month). Remove entry barriers. High-volume inflow channels
StunningBiz AI Business	SaaS 구독 (어카운트 매니저)	월 30만원 (연 360만)	수출 성장 기업 (연 수출 1~10억)	기업 RAG 맞춤 온보딩 포함. 음성 통역 무제한. 주력 매출원(전체의 50%)
StunningBiz AI Enterprise	SaaS + 온프레미스 하이브리드	월 50만원+ (연 600만+)	연 수출 10억+ 중견기업, 글로벌 기업 한국 법인	전담 CSM 배정. 보안 강화(온프레미스). 커스텀 모델. 높은 Lock-in·LTV
Regulatory certification package (premium)	Per-use billing (standalone or SaaS add-on)	CE 300,000 to 1 million eCTD 1 to 5 million	Companies pursuing regulatory approval (health supplements, cosmetics, electronics)	Exclusive access to StunningBox with 500,000 corpora. 50-70% less compared to existing RA agencies. High margin 80%+
API B2B2B	API call usage-based pricing	500,000 KRW base + 20 to 100 KRW per call	Cafe24·Naver·Amazon·CRM·ERP operators	Viral expansion as an app within the platform. Revenue proportional to user growth. Network effect

(3) Market Entry Strategy – Triple Channel Simultaneous Deployment (Securing 150 companies within 1 year of commercialization)

Channel	Execution Plan	Target	Goal to secure within 1 year
① 정부 지원사업	AI바우처 공급기관 등록(최대 4,000만원/사 무상). 수출바우처·글로벌진	수출 초보 기업. 정부 지원으로 초기 비	AI바우처 20사 + 수출바우처 10사 = 30사

	Link with export support projects. KOTRA overseas trade mission local experience	Zero cost	
② Industry organization B2B	MOU with Trade Association·KCCI·Industry-specific Export Cooperation. 20% group discount. Free seminars·Demo days once a month. Utilize references from 3 demand companies	Export growing companies. Ensure trust through association recommendation	15 companies through associations + 5 directly = 20 companies
③ Platform B2B2B	Registration in Cafe24·Naver Smart Store·Amazon Seller Central App Store. Viral spread with 'automated translation' app. 1-Click installation. Large inflow through Free Tier	E-commerce sellers (direct overseas sales). Minimizing entry barriers as an app	Including Free Tier, 100 companies (Paid conversion target of 30%+)

※ Goal to secure 150 companies including Free Tier and 50+ paid conversions through three channels within 1 year of commercialization. Achieve BEP by second year (at 300 company subscription point). The regulatory certification package is a non-competitive area based on StunningBox's exclusive data, serving as a high-margin (80%+) premium source of income.

2) Commercialization Strategy (Clarity of Business Model (BM), Marketing and Sales Channel Strategy, etc.)

(1) Business Model (BM) Structure — Quadruple Billing System and Strategic Roles

Revenue Model	Billing Method	Price Range	Expected Margin	Sales Contribution (%)	Customer Transition Path	Strategic Role
BM1. SaaS Subscription	Monthly fixed + exceeded usage charges	100,000 to 500,000 KRW/month	55~65%	50%	Free Tier → Starter → Business → Enterprise	Stable recurring revenue (ARR). Growth engine. Churn rate < 5%/month managed
BM2. Regulatory Certification Package	Per-use billing (document type×country)	CE 300,000 to 1 million eCTD 1 to 5 million	80%+	25%	SaaS customer Upselling + direct sales	Exclusive high-margin cash cow. Entry barrier through StunningBox with 500,000 corpora
BM3. RAG Custom Onboarding	Implementation cost + monthly maintenance fee	Implementation 1 to 5 million maintenance monthly 100,000 to 300,000	60~70%	15%	Enterprise customers target consulting sales	High conversion barriers (Lock-in). Average contract 24 months+. High LTV
BM4. API B2B2B	API call usage-based pricing	500,000 KRW base + 20 to 100 KRW per call	70~75%	10%	App store installation → automatic billing	Platform network Effect. Growth proportional to usage. Viral expansion

(2) Core Indicators of Profitability (Unit Economics)

Indicator	Year 1 Target	Year 3 Target	Basis of Calculation
CAC (고객획득비용)	50만원/사	30만원/사	1년차: 정부채널 무상 유입 비중 높음. 3년차: 바이럴+앱스토어 자연 유입 증가
ARPU (객단가/월)	30만원	45만원	규제인증 Upselling + Enterprise 비중 증가로 ARPU 자연 상승
LTV (고객생애가치)	720만원 (24개월)	1,620만원 (36개월)	해지율 월5%→3%로 개선, 평균 계약 기간 연장. ARPU 상승 반영
LTV / CAC 비율	14.4배	54배	SaaS 업계 건전성 기준 3배 이상 → 초기부터 대폭 초과 달성 전망
BEP (손익분기점)	2년차 말 (구독 300사)	-	월 고정비 약 1.5억원 기준. MRR 1.5억 = 구독 300사(ARPU 50만 가중평균)

(3) Marketing Strategy — Stage-wise Investment Allocation and Key Tactics

Marketing Domain	Investment Proportion	Years 1-2 (Market Penetration Phase)	Year 3 and beyond (Efficiency Phase)
Content Marketing	30%	Blogs and YouTube targeting export practitioners: practical content such as 'Consult with Buyers Without Interpreters', 'Cut Regulatory Certification Costs by 50%'. Three case studies of demand companies (Case	SEO Assetization → Over 50% Organic Inflow. Accumulation of 30+ Industry Success Stories. Regular webinars twice a month

		videos/articles production	
Performance Advertising	25%	Focus on keywords for Google Ads and Naver SA such as 'Export Translation', 'Interpretation AI'. Retargeting (encourage conversion from Free Tier users to paid). LinkedIn B2B lead generation	Concentration on channels achieving over 300% ROAS. Maintain CAC below 300,000 won. AI-based bid automation
Partner/ Channel Sales	30%	MOU with 20 KOTRA overseas trade centers → local export companies experience. Regular seminars at Trade Association. Nationwide demo tour at 62 export support centers. Joint promotions with Cafe24 and Naver	Secure 10 reseller partners by industry. Partner revenue commission model. Partnerships with overseas CRO and RA agencies
PR/Branding	15%	KOTRA's news release on export success stories. AI conference (ACL, IWSLT) paper presentations. Open-source model release → Viral marketing in the HuggingFace community	Establish 'K-AI Export Solution' brand positioning. PR in international media (e.g., TechCrunch Asia)

(4) Market Strategy — Online, Offline, Global Triad

Channel	Detailed Execution Plan	Target Number of Customers	Revenue Contribution Proportion
온라인 직판	stunningbiz.ai 웹사이트 직접 가입. Free Tier 무료 체험 → 셀프서비스 유료 전환. 카드 결제·세금계산서 자동 발행. 24시간 챗봇 CS	3년차 250사	40%
앱스토어 연동	Cafe24 앱스토어 · 네이버 스마트스토어 앱 · Amazon 셀러센트럴 플러그인. 1-Click 설치로 진입 장벽 최소화. 플랫폼 내 결제·정산 자동	3년차 150사	25%
B2B 직접 영업	Enterprise 대상 전담 영업팀(3명). 업종 협회·수출조합 연계 단체 계약. 수요기업 레퍼런스 기반 동종 업체 영업. 규제인증 패키지 컨설팅 세일즈	3년차 80사	25%
글로벌 채널	일본: 현지 IT 디스트리뷰터 파트너십. 동남아: KOTRA 해외무역관 연계 현지 체험 프로그램. 중동: 할랄·이슬람 규제 전문 RA대행사 제휴. API 수출(OEM)	4년차 200사+	10%→45%

※ Core Market Strategy: Achieve high-volume low-cost inflow via online direct sales + app store during Years 1-2 → Expand high-value Enterprise and regulatory certification sales through B2B direct sales in Year 3 → Achieve 45%+ overseas revenue contribution in Year 4 with global channels. Optimize overall marketing cost from 35% of sales (Years 1-2) to 20% (Year 3 and beyond) as content assets and partner channels become the organic growth engine.

3) Disclosure Level and Diffusion Strategy of Development Results

To balance the public interest and commercial profitability of government R&D outcomes, this project adopts an 'Open Core' strategy. AI models and datasets are made publicly available to enable free use by SMEs and research communities, while core business logic (Regulatory Certification Engine, HITL Verification, Billing) is provided only as SaaS to secure commercialization profits.

(1) Disclosure Level by Development Result — Open Core Strategy

Development Result	Disclosure Level	Disclosure Method and Utilization Plan	Reason for Non-disclosure / Commercialization linkage
Business Translation AI Model Weights	Full Disclosure	Published on HuggingFace Hub under Apache 2.0 license. Includes industry-specific variant models (Manufacturing/Retail/Food/IT). SMEs can deploy directly on their servers for free utilization	Lack of RAG, RLHF, Monitoring in Self-operation vs. SaaS → Maintain SaaS Value-add

E2E Speech Interpretation Model Weights	Full Disclosure	Two-way Streaming Interpretation Model for Korean ↔ 8 Languages. Inference Code + Docker Image Release. First Domestic E2E Korean Simultaneous Interpretation Open Source	Contribute to Academic Community + Strengthen K-AI Brand. SaaS differentiated with Speaker Separation, RAG, Plugins
Business Parallel Corpus (Training Data)	Maximum Disclosure	Business Parallel Corpus of more than 100,000 sentence pairs published on AI Hub. Full disclosure of industry-specific trade terminology dictionaries (3 types). De-identified data completed	100,000 out of 1.1 million disclosed (9%). The rest retained as non-disclosed to maintain SaaS model quality differentiation
RAG Search Engine · API Integration Code	Selective Disclosure	[Disclosure] RAG Vector Search Engine Core, Example API Integration Code (Python, Node.js), Docker Distribution Image, Developer Documentation. GitHub Open Source Repository Maintenance	[Non-disclosure] Hybrid Reranker Weights, Auto-boarding Logic by Company RAG → SaaS Key Differentiation Factors
Regulatory Certification Formatting Engine (eCTD, CE, FCC)	Non-disclosure	Provided only as a premium SaaS service. Exclusive engine based on 500,000 Stuttering Box Corpus. Accessible only via API (charged per action)	Core High-margin (80%+) Commercialization Area. Block Competition Entry. Protect 10 years of Accumulated Data Assets
HITL 검증 · 과금 · 대시보드	비공개	SaaS 플랫폼 내에서만 제공. 품질 모니터링, A/B 테스트, 자동 재학습 파이프라인, 사용량 과금 시스템	SaaS 운영 핵심 인프라. 비공개가 수익 모델의 근간

(2) Prior Establishment of Rights Relationships for Data and AI Results Origin data, fine-tuning models, and common platform rights are divided into three, and the method of transferring model weight ownership to demand companies is not adopted in accordance with public R&D result management principles.

(2-1) Ownership Plan by Deliverable

Deliverables	Attribution Entity	Content
원천 데이터	수요기업 귀속	수요기업이 제공한 원천 데이터(문서·음성·상담이력 등)
파인튜닝 모델(기업 전용)	공동 성과물	1) 수요기업 전용 가중치는 공동성과로 분류하되, 수요기업은 사업목적 범위 내 무상 사용권 보유
공통 플랫폼·베이스모델·API	주관 공동기관 귀속	통합 플랫폼 공통 인프라 및 베이스 모델은 주관·공동연구기관에 귀속

(2-2) Assured Provisions to Reflect in Contract Annexes

Attribution Entity	Content
Data provided by the demand enterprise may only be used within the scope of this business's purpose, Provision Purpose/Usage Scope Prior written consent is required for reuse or secondary learning Retention period, specify retention period and disposal procedure after project completion in the Data Provision Assurance Agreement	
Usage Rights Granted The fine-tuning results can be used freely by the demand enterprise within the scope of this business purpose Deletion-Relearning Procedure Define in advance data deletion procedures and relearning reflection procedures upon the demand enterprise's request	

(3) Outcome Diffusion Strategy — 4 Main Channels

Diffusion Channel	Detailed Execution Plan	Goals (within 3 years)	Expected Effect
Open Source Community	HuggingFace Hub model released + GitHub code release. Model card, benchmark, usage guide provided. Community operation for issue management, PR review. Quarterly releases	GitHub Star 1,000+, downloads 5,000+/month, 20 contributors+	Establishment of K-AI brand. Contributions to voluntary improvements by subsequent researchers and enterprises. Increase in SaaS reputation
Public Data Contribution	Register 100,000+ sentence pairs of business parallel corpora on AI Hub. Release 3 terminology dictionaries by industry. Obtain NIA data quality certification. Available for use as subsequent R&D training data	AI Hub downloads 3,000+ cases, utilization papers 10+	Revitalization of domestic NLP research. Solving the problem of lacking translation data for SMEs. Returning government R&D outcomes
Academic·Technical Presentations	Publication of papers at top AI conferences like ACL, EMNLP, and IWSLT (2+ papers). Publication at domestic conferences (KCC, Korean Information Science Society). 3 patent applications+(RAG, E2E interpretation, regulated formats). Technical reports available	2+ papers, 3 patents+, and 2 technical reports	Verification of technical reliability. Connection to subsequent R&D. Evidence of TRL improvement. Global technology leadership

	Public Disclosure of Reports		
Government·Institutional Related Diffusion	Distribution to 20 KOTRA overseas trade centers (free experience). Registration as AI voucher suppliers (up to 40 million won/company free). Distribution of demos to 62 trading associations and export support centers. Register excellent case studies with the Ministry of SMEs and Startups	AI Voucher 50+ companies, 200 KOTRA experiences 62 export support centers distribution	Core channel for diffusion to 500+ SMEs Key channel for distribution. Connection to subsequent government support projects. Expansion of export SME base

(4) Free Tier Continuous Operation — Guarantee Access for SMEs

Item	Contents of Free Tier Offerings	Expansion Benefit Conditions
Basic Free Offering	100 translations per month + 60 minutes of voice interpretation + 5 automatic document generations. 1GB of enterprise RAG DB. 1,000 API calls/ month. No time limit (permanently free)	30-day free trial added upon paid transition
Expanded Benefits for SMEs	Enterprises with a confirmation of SME: Double the free limit (200 uses/month+120 minutes). Up to 40 million won of free usage when linked to AI vouchers. 50% discount for first-year export beginner companies	Submission of SME certificate or export performance certificate
Free Provision for Research·Education	Unlimited free API for universities, research institutes, non-profit organizations. Separate educational dashboard provided. Joint improvement agreement for the model in joint research	Submission of certificate for educational/research purpose. Prohibition of commercial resale

※ Core strategy for public disclosure: 'Open the model, close the service' — Ensure accessibility for SMEs and community trust through open-source disclosure of AI models and datasets, while maintaining profitability by keeping core SaaS differentiation elements such as regulation certification engines, RAG automation, HITL quality management proprietary. This strategy follows the verified BM model of global open-core companies like Red Hat, Elastic, and MongoDB.

4) Expected Effects (growth potential of target market, entry feasibility, projected sales, export substitution, etc.)

(1) Growth Potential of Target Market

Market Segmentation	'24 Scale	'27 Outlook	'30 Outlook	CAGR	Source
Global AI Translation Market	1.8 billion dollars	3.5 billion dollars	7 billion dollars	30%+	Grand View Research 2024
Global Real-time Interpretation Market	500 million dollars	1.2 billion dollars	2.8 billion dollars	33%+	MarketsandMarkets 2024
Domestic AI Translation Market	120 billion won	280 billion won	650 billion won	32%+	KISTI·IITP estimation
Number of Domestic Export SMEs (Potential Clients)	About 100,000 companies	–	–	–	KOTRA Export Enterprise DB

→ The global AI translation and interpretation market is experiencing ultra-rapid growth with CAGR of 30%+, while the domestic market is expected to reach 650 billion won by 2030. A 1% market share (1,000 companies) among the potential 100,000 clients can achieve annual revenue of 22 billion won

(2) Market Entry Feasibility — Basis for Competitive Advantage

Basis for Entry	Specific Details	Level of Advantage Compared to Competitors
Monopoly Data Asset	110 million+ parallel corpora in Stunning Box (50 million+ regulated 60 million business). Impossible for competitors to replicate the exclusive data and professional translation asset accumulated over 15 years in a short time	DeepL·Google: No business-specific corpora. Domestic competitors: No regulated corpora. → 1~2 years of entry barriers
E2E Technology Preemption	Based on InfiniSST, E2E real-time interpretation (within 2 seconds) has no domestic commercialization examples. Existing	No domestic competitors (Blue Ocean). Google S2ST does not support Korean.

	Fundamental quality superiority over Cascade (STT→MT→TTS)	→ Technological lead for at least 1 year
Verified SaaS infrastructure	PCN with 25 years in IT services (Revenue 27 billion KRW, 170 employees). 110 billion KRW court SI order. Operating the Oz-i AI platform in a commercial setting. Holds GS-certified API platform.	Startup competitors: Lack of SaaS operational performance, unverified stability. → Overwhelming enterprise credibility
Securing government channels	AI voucher (up to 40 million KRW/company free) + KOTRA 20 overseas trade centers + Trade Association + 62 Export Support Centers. Secured references from 3 client companies	Minimization of initial customer acquisition cost (CAC). Possible to bring in 50 companies+ annually through government channels for free → Basis for break-even point (BEP) in 2nd year

(3) Projected Revenue and Job Creation

Classification	'28 (1st Year)	'29 (2nd Year)	'30 (3rd Year)	'31 (4th Year)	'32 (5th Year)
내수 매출	40억	60억	80억	100억	120억
수출 매출 (API·해외 SaaS)	20억	30억	50억	80억	100억
합계 매출	60억	90억	130억	180억	220억
영업이익	9억 (15%)	20억 (22%)	36억 (28%)	58억 (32%)	77억 (35%)
누적 구독 기업 수	150사	300사	500사	800사	1,000사+
누적 고용 (직접)	16명	25명	34명	42명	51명
5년 누적 매출					680억원

※ Basis for sales estimation: SaaS ARPU 300,000 KRW/month (weighted average), annual price increase of 20%, improvement of churn rate from 5% to 3% monthly, regulatory certification average unit price 1.5 million KRW·1,000 cases annually+

(4) Export Replacement Amount (Technology Export + Import Replacement Effect)

Export Replacement Items	'28	'29	'30	'31	'32
해외 SaaS·API 직접 수출	20억	30억	50억	80억	100억
해외 통·번역 외주비 수입 대체	10억	20억	30억	40억	60억
수출 대체액 합계	30억	50억	80억	120억	160억
5년 누적 수출 대체액					440억원

※ Overseas SaaS·API export: Entry into Japan·Southeast Asia·Middle East + Export of OEM API. Import replacement: Replacement effect of existing overseas translation services (DeepL Pro·Specialized translation outsourcing). Basis for calculation of export replacement: Average annual translation outsourcing cost 35 million KRW per exporting company × number of replacement companies

※ Summary: This project, in a CAGR 30%+ high-growth market, possesses commercialization potential to achieve cumulative sales of 68 billion KRW, export replacement of 44 billion KRW, and direct employment of 51 people within 5 years, based on exclusive data (1.1 million corpora) + E2E technology lead + verified SaaS infrastructure, with a very high investment efficiency (sales/support fund = 34x) compared to government support of 2 billion KRW.

8. Implementation Plan for Safety and Security Measures in R&D

1) Safety Measures Implementation Plan

During the R&D execution process, adhere to safety rules regarding the use of research facilities, equipment, and computing equipment, and provide regular safety education to participating researchers. Conduct pre-checks before equipment installation and operation, and periodically identify potential risk factors such as electricity, fire, and equipment overheating. In the event of an accident, establish an emergency contact network and response procedures to enable immediate reporting and response, and systematically manage safety inspection results and training history.

2) Security Measures Implementation Plan

Sensitive information such as research data, AI training data, demand company's products, prices, contracts, and export documents is categorized and managed according to importance. Encryption is applied to data storage and transmission sections, and access is controlled to ensure that only authorized personnel can access the data through role-based access management and user authentication. Access, viewing, and modification logs are recorded and regularly inspected, and external transfer is prevented by checking the person in charge's approval and de-identification status.

3) Implementation Plan for Other Measures

Data containing personal information and corporate secrets is clarified in terms of collection purpose and usage scope, and utilized in research after de-identification and anonymization processing. Participating researchers comply with pledges on research ethics, personal information protection, and security, and establish clear rights and utilization scopes of AI training data and research outcomes in advance. Also, regularly conduct self-inspections to verify compliance with safety, security, and ethics, and continuously manage inspection results and related supporting documents.

[Attachment 1. Confirmation Form for National R&D Project Application and Execution Details

2. Plan for Utilization and Management of Research Lab Operating Expenses for Profit Organizations Self-Inspection Results and Obligatory Confirmation
3. Form for Security-Level for Application Project

[Attachment 1. Confirmation Form for National R&D Project Application and Execution Details

Confirmation Form for National R&D Project Application and Execution Details Project Name SME Technology Innovation Development (R&D) Project Number RS-2026-25606465 Project Title Integrated Platform for Automation of Global Sales, Counseling, Translation, and Documentation for SME Export Revenue Innovation Lead R&D Institution PCN Co., Ltd. In applying for the SME Technology Development Support Project, this institution submits the list of national R&D projects that are applying or executing or have executed as provided in [Attachment 1-1], confirming that the applied project is not eligible for pre-support/pre-development or duplicate support with ongoing or completed national R&D projects. Furthermore, for verification of pre-support/pre-development status, it is confirmed that if requested by the managing agency or expert, consent to inquire about pre-support/pre-development status for the projects listed in [Attachment 1-1] is given. If any false, forged, or omitted facts in listed details are confirmed, it is agreed that no objection will be made to actions such as cancellation of selection, termination of agreement, and participation restrictions in accordance with relevant laws. [Attachment 1-1] List of Implementation/Application Details of National R&D Projects July 1, 2026 Lead R&D Institution: PCN Co., Ltd. Representative: Kwang-Hyun Song Research Director: Sang-Hyun Um To the President of the Korea Technology and Information Promotion Agency for SMEs

[Attachment 1-1] List of Implementation/Application Details of National R&D Projects (All to be filled based on the date of closure of this announcement)

◦ Lead R&D Institution

Project Number	Research and Development Project Name	Lead Research and Development Institution Name / Name and Role of Research and Development Institution (Lead/Cooperative)	Research and Development Period (Participation Period)	Contents of Implementation (Subject, Objective, Method)	Central Administrative Agency (Specialized Agency)	Remarks (In Progress/ Completed/ Applying)
RS-2023-00232134	Development of Visualization and Linkage Technology for Space Object Orbit Information Based on Metaverse	PCN Co., Ltd. PCN Co., Ltd. (Lead)	2023.04.01 ~ 2026.12.31 (2023.04.01 ~ 2026.12.31)	Real-time Visualization of Space Situational Awareness / Establishment of Orbit Information Comparison, Integration, and Metaverse Linkage System / Real-time Data Collection/Comparison and AI-based Orbit Simulation Linkage	Institute for Information & Communications Technology Planning and Evaluation	In Progress
RS-2026-25514284	Development of Emotional AI Model Generation and Training Dataset Construction Technology	PCN Co., Ltd. PCN Co., Ltd. (Lead)	2026.04.01 ~ 2028.12.31 (2026.04.01 ~ 2028.12.31.)	Infrastructure for Collecting Biometric Signals and Development of Intelligent Annotation SaaS Platform	Institute for Information & Communications Technology Planning and Evaluation	In Progress
RS-2026-\n25527887	Development of Key Technology for LCI-based Digital Service Carbon Footprint Calculation and Tracking Management	EasyG Co., Ltd. PCN Co., Ltd. (Joint)	2026.04.01 ~ 2029.12.31 (2026.04.01 ~ 2029.12.31)	Digital Service Carbon Emission Management / AI-based Life Cycle Carbon Footprint Calculation and Management System Establishment / Data Collection from Heterogeneous Systems and Data Warehouse Analysis Infrastructure Construction	Institute for Information & Communications Technology Planning and Evaluation	In Progress
RS-2026-\n25504479	Development of Character and Webtoon Copyright Opt-Out Response Technology and Evaluation Model in Generative AI Image Service Environment	Seokyeong University Industry-Academia Cooperation Foundation PCN Co., Ltd. (Participant)	2026.04.01 ~ 2028.12.31 (2026.04.01 ~ 2028.12.31)	Development of Generative AI Copyright Protection / Copyright Opt-Out Application and Infringement Identification Response Technology / Development of Copyright Infringement Evaluation Algorithm based on Large-scale Webtoon Data	Korea Creative Content Agency	In Progress
RS-2026-\n25518171	Development of AI Agent-based XR Telepresence Platform for Context Awareness of Remote Multi-party Conversations	PCN Co., Ltd. PCN Co., Ltd. (Lead)	2026.04.01 ~ 2029.12.31. (2026.04.01 ~ 2029.12.31)	XR-based Remote Collaboration Platform / Construction of Multilateral Conversation Context Awareness and XR Interaction Platform / Development of Multimodal Data Synchronization and Persona-reflecting Multi-avatar Technology	Korea Creative Content Agency	In Progress
RS-2026-\n25535833	Development of Integrated Drug Investigation System Linking Dark Web and Virtual Asset Transaction Tracking	Sungkyunkwan University Industry-Academia Cooperation Foundation PCN Co., Ltd. (Participant)	2026.04.01 ~ 2028.12.31 (2026.04.01 ~ 2028.12.31)	Comprehensive Investigation System for Drug Crimes / Dark Web / Establishment of Integrated Analysis Environment for Virtual Asset Crime Data / Development of Unstructured Data Collection, Knowledge Graph Loading, and Investigator Visualization Dashboard	Science Security Promotion Center	In Progress
RS-2026-\n25561688	Drone Video Management XR Goggle System for Disaster Response	PCN Co., Ltd. PCN Co., Ltd. (Lead)	2026.05.01 ~ 2027.12.31 (2026.05.01 ~ 2027.12.31)	Commercialization of Drone Video-based XR Management / Ultra-low Latency Drone Video Visualization and AI Situation Awareness Integrated Management System / P5G-based Transmission and VLM/ LLM-based On-site Briefing Technology Application	Korea Institute for Advancement of Technology	In Progress
RS-2026-\n25567830	Development of Semiconductor PCB Industry-specific Manufacturing Problem-solving Multi AI Agent Technology	InterX Co., Ltd. PCN Co., Ltd. (Participant)	2026.04.01 ~ 2028.12.31 (2026.04.01 ~ 2028.12.31)	PCB Manufacturing Process Optimization / Advancement of AI Visual Twin and Mass Production Line Monitoring Platform Construction / Core Process Layout Design and Equipment Data Management Pipeline Construction	Korea Institute for Advancement of Technology	In Progress

RS-2024-\n00339583	Development and Demonstration of XOps-based AI Policy Simulation Platform for Addressing Social Issues in Population Decline Areas	PCN Co., Ltd. PCN Co., Ltd. (Lead)	2024.04.01 ~ 2027.12.31 (2024.04.01 ~ 2027.12.31)	Simulation for Solving Community Issues / Construction of DataOps/ MLOps-based AI Policy Simulation Platform / Development of Multi-institutional Data Linking and Geospatial Information-based Change Monitoring UI	Institute for Information & Communications Technology Planning and Evaluation	In Progress
RS-2024-\n00354327	Development of High-Quality Ultra-Low Latency XR Cloud Computing Technology	PCN Co., Ltd. PCN Co., Ltd. (Lead)	2024.04.01 ~ 2026.12.31 (2024.04.01 ~ 2026.12.31)	Rendering Server-based XR Content / Development of High-Resolution XR Content Offloading Technology in Cloud-Native Environment / Development of HW-accelerated Rendering and Pre-warming/Load Balancing Technology	Institute for Information & Communications Technology Planning and Evaluation	In Progress
RS-2024-\n00460980	Development and Demonstration of Distributed Data Federated Computing Technology for Enhanced Data Utilization	Korea Institute of Science and Technology PCN Co., Ltd. (Participant)	2024.07.01 ~ 2028.12.31 (2024.07.01 ~ 2028.12.31)	Enhanced Data Security Federated Computing / Establishment of FedAI Central Server and Neural Network to Solve Data Silos / Design of Federated Learning Orchestration and Data Non-sensitization Module Development	Institute for Information & Communications Technology Planning and Evaluation	In Progress
RS-2024-\n00442513	Development of Workplace-contextual Understanding and Inference Capability Giant AI Model-based Robot Manipulation Task Learning Technology	Sejong University Industry-Academia Cooperation Foundation PCN Co., Ltd. (Participant)	2024.07.01 ~ 2028.03.31 (2024.07.01 ~ 2028.03.31)	AI Model for Robot Manipulation Learning / Development of VLM-based Robot Task Cognition and Synthetic Data Generation Platform / Automatic Generation and Learning Simulation of Random Synthetic Data based on Digital Twin	Korea Institute for Advancement of Technology	In Progress
RS-2024-\n00507739	AI-based Equipment Autonomous Control and State Measurement Technology Development for Semiconductor PCB Business Process Optimization	BlueNex Co., Ltd. PCN Co., Ltd. (Participant)	2024.09.01 ~ 2028.12.31 (2024.07.01 ~ 2028.03.31)	Autonomous Control of PCB Equipment / Development of AAS-based Process Recipe Autonomous Control and Quality Prediction Platform / Design of Digital Twin for Demand Companies and Connection of Physical Operation Simulation	Korea Institute for Advancement of Technology	In Progress
RS-2025-\n25449059	Development of Multi-purpose Autonomous Orchard-Agricultural Work Robot Partnering Platform for Innovation in Future Agricultural Productivity	PCN Co., Ltd. PCN Co., Ltd. (Lead)	2025.09.01 ~ 2028.12.31 (2025.09.01 ~ 2028.12.31)	Comprehensive Management of Autonomous Agricultural Work Robots / Demonstration of HAARS Platform for Autonomous Orchard and Agricultural Work Robots / Construction of Data-driven AI Decision Support and Robot Partnering Marketplace	Korea Institute for Advancement of Technology	In Progress
RS-2026-\n25516382	Development of Quantitative Evaluation Technology for Emotional and Sensory Reactions to Virtual Convergence Content	Korea Institute of Science and Technology PCN Co., Ltd. (Participant)	2026.04.01 ~ 2028.12.31. (2026.04.01 ~ 2028.12.31.)	Emotion Evaluation in XR Content / Development of Multimodal-based Emotion Measurement and Evaluation Indicators in XR Content / Development of Biomonitoring Patch and AI-based Emotion Recognition Evaluation Model	Institute for Information & Communications Technology Planning and Evaluation	In Progress
RS-2026-\n25522497	Development of Multi-party Immersive Experience Visualization Space-Audio Composition and Reproduction Rendering Technology	Korea Electronics Technology Institute PCN Co., Ltd. (Participant)	2026.04.01 ~ 2028.12.31 (2026.04.01 ~ 2028.12.31)	Reproduction of Space-Audio for Immersive Experience / Development of 3D Virtual Space and Custom Audio Rendering based on Multimodal Data / Data Collection and Place Attribute Inference using VQA-VLM based on Geospatial Information	Institute for Information & Communications Technology Planning and Evaluation	In Progress

[Appendix 2. Plan for Utilization and Management of Research Laboratory Operation Costs of Profit Organizations Not Applicable]

영리기관의 연구실운영비 활용·관리 계획

1. Status of R&D Project

Project Title				
Research and Development Project Name				
Research and Development Institution Name				
Research Director	Affiliation		Name	
Research and Development Period	Total	20 . . . ~ 20 . . . (year month)		
	First Year	20 . . . ~ 20 . . . (year month)		
	nth Year	20 . . . ~ 20 . . . (year month)		

2. Research Laboratory Operation Costs for Profit Organizations (Amount)\n(Unit: Thousand KRW)

Classification	Consumable Costs Necessary for Laboratory Operation	Costs for Office Equipment and Software (A)	Costs for Equipment and Supplies to Maintain a Comfortable and Clean Environment in the Laboratory (B)	Total\n(C=A+B)
Overall Period	(Not Available)			
Stage 1	(Not Available)			
nth Stage	(Not Available)			

* Enter the amount planned at the time of the initial research and development project agreement signing\n** Enter costs for purchasing, installing, leasing, and using office equipment and software in A\n*** Enter costs for purchasing and maintaining equipment and supplies necessary to maintain a comfortable and clean laboratory environment in B\n**** Add or delete rows according to the annual structure of the research and development project

3. Research Laboratory Operation Costs for Profit Organizations (Items)\n(Unit: ea, Thousand KRW)

Classification		Classification	Quantity (G)	Unit Price (H)	Amount (I=G×H)
First Year					
	Item Name				
	Item Name				
nth Year					
	Item Name				

* Enter the amount planned at the time of the initial R&D project agreement ** Enter the item name of the equipment, software, or supplies in 'Item Name' *** Enter 'Office Use' in the 'Classification' field when using (planning) costs for office equipment and software, or 'Environmental Maintenance' when using (planning) costs for equipment and supplies for maintaining heating, cooling, and cleanliness in the laboratory **** Rows can be added or deleted according to the stage configuration of the R&D project ***** The content entered in Article 3 must be consistent with the content entered in Article 2 ****** For-profit institutions must submit this document to the head of the central administrative agency when applying for an R&D project (If not submitted, retrospective application is not possible, and if the laboratory operation cost is not intended to be used, it is excluded)

Year Month Day

Research Manager: ㅇ ㅇ ㅇ (Seal) Main/Joint/Subcontracted
Research and Development Institution: ㅇ ㅇ ㅇ (Seal)

Director of Small and Medium Business Technology Information Promotion Agency

[Attachment 3. Self-inspection results of the security grade of the applying project and obligation confirmation upon designation as a security project

Not Applicable

Self-inspection results of the security grade of the applying project and upon designation as a security project

Obligation Confirmation

Project Name			
Project Number			
Project Name			
Main Research and Development Institution		Main Research Manager	
Joint Research and Development Institution		Joint Research Manager	
Subcontracted Research and Development Institution		Subcontracted Research Manager	

Our institution confirms that upon applying for the above-mentioned task in the Small and Medium Business Technology Development Support Project, the security grade matches the attached document .

Our institution has confirmed all the attached security measures for the Small and Medium Business Technology Development Support Project hereinafter referred to as 'Security Measures', and pledges that if the project applied for is confirmed as a security project, we will perform the duties and responsibilities as a research and development institution conducting security tasks stipulated in the Security Measures

In addition, if the security grade of the above-mentioned project is designated as a 'security project' by the classification results of the security task classification committee of the professional institution during the execution of the above project, or due to changes in government policy, while it was applied as a 'general task' based on the self-inspection results, we will fulfill the duties and responsibilities as a research and development institution conducting security tasks stipulated in the Security Measures .

Our institution pledges to bear sanctions and civil and criminal liability according to the National Research and Development Innovation Act and related security law provisions if we violate obligations according to the Security Measures .

<Attached Scheduled Task Security Grade Classification Self-Inspection Table .

Small and Medium Business Technology Development Support Project
Security Measures

20 Year Month Day

Main Research and Development Institution Name: Representative: (Seal) Manager: (Seal) Joint Research and Development Institution Name: Representative: (Seal) Manager: (Seal) Subcontracted Research and Development Institution Name: Representative: (Seal) Manager: (Seal)

Minister of Small and Medium Enterprises and Startups (Director of Small and Medium Business Technology Information Promotion Agency) ( )

<Attachment 1> Not Applicable

Scheduled Task Security Grade Classification Self-Inspection Table

□ Based on Article 45 of the Enforcement Decree of the National Research and Development Innovation Act (classification of security projects related to research and development tasks), please inspect. (Security grades are 'security project' and 'general project'.) □ Tasks classified as security projects must be managed according to <Attachment 2> 'Small and Medium Business Technology Development Support Project Security Measures' and the internal security management regulations of the research and development institution conducting the security project.

Project Title			
Project Number		Inspection Date	20 . . .
Project Name	Transcribe the project name entered in the project reception system as is		
Main Research and Development Institution		Main Research Manager	
Joint Research and Development Institution		Joint Research Manager	
Subcontracted Research and Development Institution		Subcontracted Research Manager	

Number	Security Grade Classification and Assessment Criteria	Assessment Result	
		Yes	No
1	Research and development project related to the defense improvement project pursuant to Article 3(1) of the Defense Business Act		
2	Research and development project related to technology under nationalization following refusal of technology transfer abroad		
3	Research and development project related to future core technology recognized by the head of the central administrative agency as needing protection		
4	Research and development project related to national core technology pursuant to Article 2(2) of the Act on Prevention and Protection of Industrial Technology Leakage		
5	Research and development project related to technology requiring export permission or restriction under Article 19(1) of the Foreign Trade Act		
6	Tasks recognized as needing to be classified as security projects for other reasons		
※ If the research institution applies as a 'security project', detailed reasons must be provided			

Security Grade Final Confirmation Result: □ Security Project □ General Project ※ If there is any 'Yes' in the above review results, it is classified as a security project

Our institution confirms that we have sincerely reviewed and completed all the above items.

20 Year Month Day

Main Research and Development Institution Name: Representative: (Seal) Manager: (Seal) Joint Research and Development Institution Name: Representative: (Seal) Manager: (Seal) Subcontracted Research and Development Institution Name: Representative: (Seal) Manager: (Seal)